ESSENTIAL MATHEMATICS GRADE 10: AREA OF POLYGONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.4: Area of Polygons
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Area of a triangle given two sides and an included acute angle (formula: ½ab sin C) – Area of a triangle given three sides using Heron's Formula ($s = (a+b+c)/2$; $A = \sqrt{[s(s-a)(s-b)(s-c)]}$) – Area of a rhombus (using diagonals: $A = ½d_1d_2$; and using base and height: A = base × height) – Area of a parallelogram (A = base × height; $A = ab \sin \theta$) – Area of a regular pentagon ($A = (5s^2)/(4 \tan 36°)$) – Area of a regular hexagon ($A = (3\sqrt{3}/2)s^2$) – Applications of areas of polygons in real-life situations (floor tiling, fabric patterns, land surveying, construction)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) compute area of a triangle given two sides and an included acute angle, b) determine the area of a triangle given three sides using Heron's Formula, c) determine the area of a rhombus and parallelogram in different situations, d) determine the area of a regular pentagon and hexagon from real life situations, e) appreciate the application of areas of polygons in real-life situations.
Core Competencies	– Creativity and Imagination: The learner acquires networking skills by undertaking group activities and exchanging new ideas while discussing the applications of area of polygons in contexts such as Kenyan kanga fabric patterns, floor tiles, and maize-farm layouts. – Learning to Learn: The learner learns independently when measuring the sides of a triangle and works out the area using Heron's Formula, developing self-directed inquiry habits as they move from familiar triangular shapes (tripod stands, roof trusses) to formal algebraic procedures.
Core Values	– Unity: The learner shares available resources amicably while using selected objects with the shape of regular pentagons and hexagons when working out the area, demonstrating fairness and team spirit during group measurement tasks. – Responsibility: The learner offers leadership to others while discussing with peers the applications of area of polygons, taking ownership of group outcomes and ensuring geometric instruments (rulers, geometrical sets) are handled with care and returned in good condition.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners articulate why different polygon formulas are needed by questioning how to find the area of a triangular maize plot when no perpendicular height is measurable, motivating the need for the sine rule

	and Heron's Formula. – Planning and Carrying Out Investigations: Learners measure physical objects (kanga fabric tiles, classroom floor tiles, tripod-stand tops, hexagonal paving stones) and apply appropriate polygon-area formulas, comparing computed areas with direct estimates. – Using Mathematics and Computational Thinking: Learners select and apply the correct area formula (½ab sin C, Heron's, rhombus diagonal, regular polygon) for each polygon type, checking dimensional consistency and reasonableness of answers. – Constructing Explanations: Learners construct written and oral explanations linking the formula used to the geometric properties of each polygon (e.g., why the sine of the included angle appears in the triangle formula). – Obtaining, Evaluating, and Communicating Information: Learners present their area calculations and real-life findings to peers, evaluating the accuracy and relevance of each other's measurements and formula choices.
Pertinent & Contemporary Issues (PCIs)	– Social Awareness Skills: The learner develops effective communication skills when discussing with peers the applications of area of polygons, learning to articulate mathematical reasoning clearly and respectfully in group settings. – Education for Sustainable Development: By computing areas of agricultural plots, tiling surfaces, and community structures using locally available polygon-shaped objects, learners appreciate how mathematical skills support sustainable land use and resource-efficient construction in Kenyan communities.
Career Connections	– Land Surveying and Cartography: Area-of-polygon calculations underpin the measurement and subdivision of land parcels in Kenya's counties — a direct skill used by land surveyors and GIS technicians working under the National Land Commission. – Architecture and Building Construction: Calculating floor areas of rooms, roof sections, and decorative tiling patterns (triangular, hexagonal) is a daily task for architects, quantity surveyors, and building technologists on Kenyan construction sites. – Textile and Fashion Design: Kenyan fabric designers and tailors use polygon-area calculations to estimate cloth requirements for kanga, kitenge, and school-uniform cuts, minimising material waste and controlling production costs. – Agriculture and Farm Management: Farmers in the Rift Valley and Central Kenya compute areas of polygonal plots to determine seed quantities, fertiliser rates, and expected yields — skills supported by area-of-polygon knowledge. – Sports Coaching and Facilities Management: Coaches and grounds managers calculate areas of triangular penalty zones, hexagonal court markings, and parallelogram-shaped athletic tracks to plan maintenance and equipment use. – Journalism and Photography (Media): Graphic designers and photojournalists use polygon-area principles when compositing images, laying out print pages, and computing proportional scaling of infographics.
Focus for Lessons	This unit develops learners' ability to compute areas of triangles (using the sine formula and Heron's Formula), rhombuses, parallelograms, regular pentagons, and regular hexagons, moving from familiar shapes in the Kenyan environment to formal algebraic generalisations. Throughout the unit, learners engage in hands-on measurement of real objects — floor tiles, fabric panels, farm plots, and tripod-stand tops — before applying and connecting multiple area formulas. The pedagogical approach centres on collaborative investigation, guided sense-making, and progressive model-building anchored in the phenomenon of why the same piece of land or fabric can be measured in different ways depending on the information available.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its height — so that builders, farmers, and designers in our community never order too much or too little material? KICD KEY INQUIRY QUESTION: 1. How are areas of polygons useful?
Anchoring Phenomenon	A community in Kericho County needs to pave the courtyard of a new community hall with decorative tiles. The courtyard floor is not a simple rectangle — it is divided by chalk lines into triangular sections, a large parallelogram stage area, and a border of

	hexagonal stone tiles. The building contractor has measurements of some sides and angles but NOT the perpendicular heights of most sections, and the hexagonal border tiles come in only one size. Learners observe photographs and a scale drawing of the courtyard and are confronted with the puzzle: the contractor knows the lengths of two sides of each triangular section and the angle between them — but the usual "base × height ÷ 2" formula cannot be applied directly because no right angle exists and no height can be physically measured. Similarly, for the hexagonal border tiles, the contractor only knows the side length of each tile. How can the total area be calculated accurately enough to order the correct amount of material — without being able to drop a perpendicular? This is genuinely puzzling because learners' prior experience of area always relied on a visible height, yet here the heights are inaccessible. The phenomenon anchors the entire unit: every formula developed (sine-area, Heron's, parallelogram, pentagon, hexagon) is a tool that solves a real measurement problem in the Kericho courtyard scenario.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Predict Phase (Triangle: ½ab sin C introduced): Learners are shown the Kericho courtyard scale drawing and photographs. In pairs they predict how much tiling material is needed, using only "base × height ÷ 2" — and discover they are stuck because no heights are given. The teacher cold-calls 4 pairs to share predictions and records the range on the board. The class generates initial questions for the Driving Question Board (DQB). The teacher then guides learners to re-examine a right-angled triangle, derive that height = b sin C, substitute into ½ × base × height, and arrive at A = ½ab sin C. Learners apply the formula to two triangular sections of the courtyard drawing. Lesson 2 — Observe & Explain Phase (Triangle: ½ab sin C practised; Heron's Formula introduced): Learners measure two sides and the included angle of triangular objects in the classroom (set-square faces, corner of a book cover cut diagonally) and compute areas, comparing with estimates. The teacher introduces the Rift Valley farm sub-phenomenon: three sides known, no angle given. Learners are guided to derive Heron's Formula from the sine-area formula through a structured worked example. Small groups work out the area of the triangular maize plot. The DQB is updated: "What if we only know all three sides?" Lesson 3 — Deep Practice & Model Building Phase (Heron's Formula consolidated): Learners work in groups to identify triangular shapes from the immediate environment (tops of tripod stands, triangular roof sections visible from the window). They measure three sides, apply Heron's Formula, and record results in a class table. A whole-class gallery walk compares computed areas. Learners revise their initial courtyard area model to include Heron's Formula for the triangular sections. Misconception check: teacher asks, "Does it matter which side you call a, b, or c in Heron's Formula?" — 3-minute think-pair-share, then cold-call. Lesson 4 — Observe & Explain Phase (Area of Rhombus and Parallelogram): Learners bring or identify objects with rhombus/parallelogram shapes (kanga fabric patches, floor-tile arrangements, parallelogram-shaped card offcuts). The textile sub-phenomenon is introduced. Learners measure diagonals and apply $A = ½d_1d_2$ for the rhombus; they measure base, side, and included angle and apply A = ab sin θ for the parallelogram. The teacher connects the sine formula back to Lesson 1, reinforcing the storyline: "We used ½ab sin C for a triangle — how is the parallelogram formula related?" Learners update the DQB. Lesson 5 — Practise & Apply Phase (Rhombus and Parallelogram in context): Learners solve a set of structured problems involving the Kericho courtyard parallelogram stage area and additional real-life contexts (a kanga patch, a school garden in parallelogram shape). They compare both formulas (diagonal method vs. sine method) and discuss when each is more practical. Peer assessment using a checklist. The courtyard model is updated with the parallelogram stage area calculated. Lesson 6 — Observe & Explain Phase (Area of Regular Pentagon): The market-vendor sub-phenomenon (pentagon njugu tray) is introduced. Learners investigate the interior angles of a regular pentagon (sum = 540°, each = 108°), divide it into 5 congruent isosceles triangles, and derive $A = (5s^2)/(4 \tan 36°)$. Guided worked example using a measured pentagon tile. Learners apply the formula to find the area of the pentagonal tray from the sub-phenomenon. The DQB is updated: "Why does tan 36° appear in the pentagon formula?"

	Lesson 7 — Observe & Explain Phase (Area of Regular Hexagon): The honeycomb sub-phenomenon is introduced. Learners examine hexagonal objects (paving stones, pencil cross-sections, honeycomb diagrams). They divide the regular hexagon into 6 equilateral triangles and derive $A = (3\sqrt{3}/2)s^2$. Learners calculate the total wax area of a honeycomb frame (given cell side length and number of cells). The teacher connects the hexagon formula to the triangle formula from Lesson 1, completing the storyline arc. DQB updated with final answers to opening questions. Lesson 8 — Synthesis, Model Building & Assessment Phase: Learners return to the full Kericho courtyard scenario. Using all formulas developed across the unit, they calculate the total area of every section (triangles by sine formula and Heron's, parallelogram stage, hexagonal border tiles) and determine the total material needed. Groups present their courtyard area calculations and justify formula choices. The teacher facilitates a whole-class model revision — learners annotate the original courtyard drawing with correct areas. A written class activity (portfolio task) assesses all five specific learning outcomes individually. The DQB is reviewed and closed, with learners articulating how each lesson's formula answered one of their original questions.
Supporting Phenomena	– A farmer in the Rift Valley owns a triangular maize plot whose three boundary lengths are known from a GPS device, but the plot is hilly and no perpendicular height can be walked out — motivating Heron's Formula as the only practical tool. – Traditional Kenyan kanga and kikoi fabrics are woven in repeating rhombus and parallelogram patterns; a textile student wants to calculate the area of each coloured patch to estimate dye quantities, but only the side lengths and the acute angle of each parallelogram patch are labelled on the design template. – Honeybee farmers in Baringo County observe that honeycomb cells are regular hexagons; they want to estimate the total wax surface area of a frame to understand honey-storage capacity — knowing only the side length of one cell and the number of cells per frame. – Nairobi street vendors arrange groundnut (njugu karanga) display trays in the shape of regular pentagons to maximise visibility from multiple directions; a market researcher needs to calculate the display area of each tray from a single side measurement to compare vendors' stall sizes.

LESSON 1 (40 minutes): Anchoring the Courtyard Puzzle — Discovering Why "Base × Height ÷ 2" Is Not Enough	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor the unit phenomenon by confronting learners with the real measurement problem faced by a Kericho building contractor, exposing the limitation of the familiar base-times-height area formula, and guiding them to derive the sine-area formula $A = \frac{1}{2}ab \sin C$ for the first time.
Knowledge	– State the standard triangle area formula $A = \frac{1}{2} \times \text{base} \times \text{height}$ and identify when it cannot be directly applied – Recognise that the perpendicular height of a triangle can be expressed as $h = b \sin C$ when two sides and their included angle are known – State and write the sine-area formula $A = \frac{1}{2}ab \sin C$ and identify the variables a, b, and C
Skills	– Derive $A = \frac{1}{2}ab \sin C$ by substituting $h = b \sin C$ into the standard formula, showing each algebraic step – Apply $A = \frac{1}{2}ab \sin C$ to calculate the area of a triangle given two sides and their included angle, using a calculator to evaluate sin C
Attitudes	– Appreciate that new mathematical tools arise from genuine real-life measurement problems faced by builders and designers in Kenyan communities – Show curiosity and persistence when a familiar method fails, embracing the challenge of finding an alternative approach
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	This opening lesson launches the Kericho courtyard phenomenon, reveals the critical gap in learners' prior knowledge (no perpendicular height is available), and equips them with the first new formula — $A = \frac{1}{2}ab \sin C$ — which they immediately apply to two triangular sections of the courtyard scale drawing, beginning the journey toward ordering the correct amount of tiling material.
Safety Notes	No specific hazards. Ensure learners handle rulers and protractors carefully when working on the scale drawing. If printed photographs are distributed, ensure no sharp edges on cut materials.

B. LESSON OVERVIEW

This is the opening lesson of Sub-Strand 2.4 and it serves as the anchoring event for the entire unit. Learners are introduced to a vivid, real-world scenario: a building contractor in Kericho County must calculate the total area of a community hall courtyard before ordering decorative tiles. The courtyard is divided into triangular sections, a parallelogram stage area, and a border of hexagonal stone tiles — none of which yield a measurable perpendicular height. The lesson deliberately places learners in a state of productive confusion: they are asked to compute area using only what they already know, and they discover, pair by pair, that they are stuck. This cognitive conflict is the engine that drives the rest of the unit.

In the first half of the lesson learners work in pairs to attempt area calculations using the scale drawing and only the measurements provided (two side lengths and one included angle per triangle — no heights). Four pairs share their predictions with the class, and the teacher records the range of estimates on the board, making visible how uncertain everyone is. The class then generates genuine questions for the Driving Question Board (DQB), which remains posted in the classroom and is updated every lesson. This public record of what learners do not yet know is a key feature of the storyline pedagogy.

In the second half of the lesson the teacher facilitates a guided re-examination of a right-angled triangle. Using a diagram projected or drawn on the board, learners are led step-by-step to see that the perpendicular height from vertex B to side AC can be written as $h = b \sin C$, and that substituting this into $A = \frac{1}{2} \times \text{base} \times \text{height}$

immediately gives A = ½ab sin C. Learners then apply this new formula to two labelled triangular sections of the Kericho courtyard scale drawing, recording their answers and comparing with neighbours. The lesson closes with an initial sketch model — a hand-drawn diagram annotating the courtyard — that learners will revise across the entire unit as each new polygon formula is developed.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive (or view on screen) a printed A4 scale drawing of the Kericho community hall courtyard and two accompanying photographs. Working in assigned pairs, they read the contractor's data card: each triangular section of the courtyard has two side lengths and the angle between them labelled, but no perpendicular height is given anywhere. Each pair attempts to estimate the total area of just the two labelled triangular sections using any method they already know, writing their working and answer on a half-sheet of paper. After 4 minutes, four pairs are cold-called to share their prediction and their method (or their reason for being stuck).	Distribute the scale drawing and photographs as learners settle. Say exactly: 'A contractor in Kericho needs to tile this courtyard. Your job right now — before any new teaching — is to use what you already know to estimate the area of Triangle A and Triangle B on your drawing. Work with your partner. You have 4 minutes.' Start a visible 4-minute timer. Circulate silently for the first 2 minutes, noting which pairs attempt base × height, which use guessing, and which write 'cannot do — no height.' After 2 minutes say: 'If you are stuck, write down exactly WHY you are stuck. That is equally valuable.' At the end of 4 minutes, cold-call exactly 4 pairs in sequence — choose one pair that attempted a calculation, one that is stuck, one that guessed, and one that articulated the problem clearly. For each pair ask: 'What did you try? What answer did you get — or what stopped you?' Record all four predictions and methods on the board in a table with columns: Pair | Method Used | Area Estimate | Problem Encountered. Allow 10 seconds of WAIT TIME after each pair speaks before moving on. Do NOT correct any answer yet. Close by saying: 'Notice we have four different answers and at least two pairs who could not even begin. That	Activating Prior Knowledge with Productive Failure — learners are deliberately asked to apply the base-times-height formula in a context where it cannot work, making the limitation vivid and personally experienced rather than simply stated. Recording the range of predictions on the board externalises the uncertainty and creates shared motivation to resolve it.	Teacher looks for: (1) Can learners correctly state the standard formula A = ½ × base × height? (2) Do learners recognise that without a height the formula cannot be directly applied? (3) Can at least one pair articulate that they need the perpendicular distance from a vertex to the opposite side? Listen for the phrase 'we don't have the height' as evidence of productive awareness. Any pair that simply multiplies the two given sides together (without ÷ 2 or × sin C) is flagged for targeted questioning in the Explain Phase.

		gap — between what we know and what we need — is exactly what this unit is going to close.'		
Observe Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The teacher projects or draws a large, clearly labelled right-angled triangle on the board — Triangle PQR with the right angle at Q. Learners observe as the teacher annotates: side PQ = a (base), side QR = b (a second side), and angle QPR = C (the angle at vertex P between sides a and b... wait — teacher then shows the non-right-angled version). The teacher then shows a second diagram: the same triangle PQR but now with NO right angle at Q. A dashed perpendicular height h is dropped from Q to PR, creating a smaller right-angled triangle inside. Learners observe as the teacher writes the relation sin C = h / b on the board step by step. They copy both diagrams and the relation into their exercise books. Learners then examine the two triangular sections of the Kericho courtyard scale drawing again, this time specifically looking for what is given (two sides and an included angle) and what is missing (the perpendicular height).	Say: 'Let us go back to something you know very well — a right-angled triangle.' Draw Triangle PQR with right angle at Q on the board, label a, b, h clearly. Ask: 'What is the area of this triangle?' Cold-call 1 learner. Confirm: 'A = ½ × a × h. Good.' Then say: 'Now watch carefully. I am going to remove the right angle.' Redraw the triangle without the right angle, keeping the same labels P, Q, R. Draw the dashed perpendicular from Q to side PR, label it h. Say: 'This height h is now inside the triangle — we cannot measure it physically on the courtyard floor. But look at this smaller right-angled triangle on the right.' Point to the sub-triangle. Ask: 'In this small right-angled triangle, the hypotenuse is side QR which we called b. The side opposite angle C is h. So what does trigonometry tell us about sin C?' Allow 12 seconds WAIT TIME. Cold-call 1 learner. Write on board: sin C = h / b, therefore h = b sin C. Say: 'Everyone write that in your book — this is the key step.' Give 30 seconds for writing. Then ask learners to turn back to their courtyard scale drawing: 'Look at Triangle A. Tell your partner: what two sides are labelled, and what angle is between them?' Allow 1 minute of paired talk. Cold-call 1 pair to confirm the values aloud.	Phenomenon-Anchored Diagram Analysis — the teacher uses the familiar right-angled triangle as a bridge to the non-right-angled case, then immediately connects the new relationship (h = b sin C) back to the specific triangles in the Kericho courtyard drawing. This keeps the observation grounded in the real context rather than being an abstract geometry exercise.	Teacher checks: (1) Do learners correctly copy both diagrams with labels? Scan books during the 30-second writing pause. (2) When cold-called, can a learner correctly state sin C = h/b unprompted? (3) During the paired talk about the courtyard drawing, do both partners correctly identify the two given sides and the included angle in Triangle A? If learners confuse the included angle with a non-included angle, address immediately by pointing to the vertex where the two labelled sides meet.
Explain Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane]	Building directly on the observation, learners watch and participate as the teacher completes the substitution on the board: replacing h with b sin C in	Write the substitution step by step: 'We said A = ½ × a × h. We just found h = b sin C. So let us substitute:' Write A = ½ × a × (b sin C) = ½ab sin C in large notation.	Worked Example → Immediate Application — the teacher models the full derivation and one worked example, then immediately releases responsibility to learners	Teacher checks calculators and written working during the 3-minute independent work time, specifically looking for: (1) Correct key sequence: ½ × 8 × 11 ×

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the formula A = ½ × a × h to arrive at A = ½ab sin C. Each learner writes the full derivation in their own books. The teacher then displays Worked Example 1 (Triangle A from the courtyard: sides 8 m and 11 m, included angle 65°) and Worked Example 2 (Triangle B: sides 6.5 m and 9 m, included angle 48°). Learners attempt both calculations independently using a scientific calculator, then compare answers with their neighbour before the teacher confirms the correct values on the board. Learners annotate their courtyard scale drawing with the calculated area values, writing them inside each triangle.	Underline the final formula and say: 'This is the sine-area formula. It works for ANY triangle where you know two sides and the included angle — you never need to measure the height.' Ask the whole class: 'In words, what are a, b, and C in this formula?' Cold-call 2 learners and correct any confusion (a and b are the two known sides; C is the angle between them — the included angle). Display Worked Example 1 on board: 'Triangle A: a = 8 m, b = 11 m, C = 65°. Calculate the area.' Say: 'Work this out on your own. You have 3 minutes.' Set timer. After 3 minutes, ask learners to show their calculator screens to a neighbour and compare. Cold-call 1 learner to read their answer. Write the correct working: A = ½ × 8 × 11 × sin 65° = ½ × 8 × 11 × 0.9063 = 39.88 m² ≈ 39.9 m². Repeat for Worked Example 2 (Triangle B: a = 6.5 m, b = 9 m, C = 48°): A = ½ × 6.5 × 9 × sin 48° ≈ 21.77 m² ≈ 21.8 m². Say: 'Write these areas inside the triangles on your courtyard drawing. The contractor now knows the area of two sections. How many sections remain?' Pause for 8 seconds. Learners note that the parallelogram, pentagon, and hexagonal border still need formulas — connecting forward to upcoming lessons.	for an identical parallel example set in the same Kericho phenomenon context. Annotating the scale drawing with computed values makes the mathematics spatially visible and ties every calculation to the real courtyard, reinforcing that each formula is a tool for solving the contractor's actual problem.	sin(65) — watch for learners who compute sin(65) in degrees mode vs. radians mode (a common error); if a learner gets 0.826 instead of 0.906, their calculator is in radian mode — correct this immediately. (2) Correct rounding to 1 decimal place. (3) Does the learner correctly label a and b as the two sides and C as the angle between them (not any angle of the triangle)? Exit check: ask 3 random learners to state the formula from memory without looking at the board.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan	Each learner receives two sticky notes (or tears a page into two strips). On the first they write one question they still have about calculating areas of the courtyard shapes — anything genuinely puzzling them right now. On the second they write one thing they feel confident they understand	Reveal the Driving Question Board (a large piece of manila paper or a section of the whiteboard with the unit's driving question written at the top in bold: 'How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its height — so that builders, farmers,	Driving Question Board — a public, cumulative knowledge-tracking tool rooted in the anchoring phenomenon. By having learners generate their own questions immediately after the first formula is introduced, the DQB captures authentic curiosity and creates a visual record of the	Teacher reads all posted questions before the next lesson. Look for: (1) Questions that reveal misconceptions (e.g., 'Do we always use sine?' — suggests learner may not have understood the two-sides-and-included-angle condition). (2) Questions that anticipate upcoming content

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	from today's lesson. Both strips are posted on the class Driving Question Board, which is divided into two columns: 'What We Are Still Wondering' and 'What We Are Beginning to Understand.' Learners briefly read a few of their classmates' questions posted near them.	and designers in our community never order too much or too little material?'). Say: 'This board belongs to all of us. Every lesson, we will add to it and move things from the wondering column to the understanding column. Right now, take 2 minutes to write your two strips.' Set a 2-minute timer. After posting, read 3–4 questions aloud from the 'wondering' column — prioritise questions that point forward to upcoming lessons (e.g., 'What about the hexagonal tiles — they are not triangles?', 'Can we use this formula when the angle is obtuse?', 'What is Heron's formula for when we have no angle at all?'). For each, say: 'Great question — write the lesson number next to it as we answer it together over the coming weeks.' Do NOT answer the questions now. Say: 'These questions are your map for the rest of this unit. The contractor in Kericho cannot order tiles until we answer all of them.'	unit's knowledge-building trajectory. Revisiting the board every lesson builds metacognitive awareness of growing competence.	correctly (Heron's formula, hexagon formula) — these learners may benefit from extension tasks. (3) Any learner who cannot write even one question may need a check-in at the start of Lesson 2. Photograph the DQB after class for the teacher's planning record.
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES	Each learner opens a fresh page in their exercise book titled 'My Kericho Courtyard Model — Lesson 1.' They sketch a simplified version of the courtyard layout (copying the key shapes from the scale drawing: two triangles, one parallelogram, the hexagonal border region). Inside Triangle A and Triangle B they write the computed area values. For all other shapes they write a large question mark and label the shape (e.g., 'Parallelogram — area formula?', 'Hexagonal border — formula?'). Below the sketch they write in a box: 'Formula learned today: A = ½ab sin C — use when two sides and the included angle	Say: 'In every lesson of this unit, you will update a sketch of the Kericho courtyard as you learn each new formula. This is your thinking model — it shows what you know and what you still need to find out.' Display a teacher-drawn example of the initial model on the board (simple outline, two triangles with area values, question marks on remaining shapes). Say: 'Copy this layout now. It does not need to be perfectly to scale — it just needs to show the shapes and what we know so far. You have 3 minutes.' Set timer. Circulate and check that learners are writing the formula box at the bottom. After 3 minutes,	Cumulative Annotated Diagram Model — learners build and maintain a personal visual model of the phenomenon that grows lesson by lesson. The question marks on unlearned shapes create a cognitive scaffold: learners always know exactly what they know and what remains to be learned. Connecting the model back to the opening predictions reinforces that mathematics is a tool for resolving genuine uncertainty, not an abstract exercise.	Teacher circulates during the 3-minute sketching time and checks: (1) Is the formula A = ½ab sin C written correctly (not A = ab sin C or A = ½ab cos C)? (2) Are the two computed area values (39.9 m² and 21.8 m²) correctly placed inside the right triangles? (3) Are the remaining shapes labelled with question marks rather than left blank (indicating learner awareness of what is still unknown)? Collect 4–5 books at the end of the lesson to check model quality and return at the start of Lesson 2 with brief written feedback.

for similar readings	are known.' This sketch becomes the learner's running model, updated in every lesson of the unit.	say: 'Each lesson, your question marks will be replaced with formulas and calculated areas. By the final lesson, your model will show the complete area of the whole courtyard — and the contractor in Kericho will be able to order exactly the right amount of material.' Close by revisiting the board of four predictions from the Predict Phase: 'Remember those estimates from the start of class? Now you have real calculated values. Were any of the estimates close? Next lesson, we will add another piece of the puzzle — Heron's formula — for when we know three sides but no angle at all.'		

D. TEACHER REFLECTION
1. During the Predict Phase, how many pairs were genuinely stuck (could not write any working at all), and does this suggest learners' prior trigonometry knowledge from Sub-Strand 2.3 is strong enough to support the sine-area derivation in Lesson 2 and beyond — or do I need to build in a brief trigonometry recap at the start of Lesson 2? 2. When I cold-called learners to identify a and b and C in the formula, were there consistent errors (e.g., choosing a non-included angle)? If so, how will I address the "included angle" concept more explicitly in Lesson 2 when Heron's formula is introduced as an alternative for when no angle is known? 3. Did all learners correctly set their calculators to degree mode? How many got wrong answers for sin 65° or sin 48°? Do I need to build a 2-minute calculator-mode check into the opening of every lesson this unit? 4. Looking at the DQB questions posted today, which questions point most directly to the next three lessons (Heron's formula, rhombus/parallelogram, pentagon/hexagon)? How can I sequence my lesson openings to explicitly pick up those questions and show learners their wondering is being answered? 5. Were the learners' sketch models complete and correctly annotated at the end of the Model Building Phase? If several models were missing the formula box or had incorrect area values, should I start Lesson 2 with a 3-minute peer-review of the model before introducing new content?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a scale drawing and photographs of the Kericho community hall courtyard, noting that each triangular section has two labelled side lengths and one included angle but no perpendicular height. They observed the contractor's dilemma: the familiar area formula A = ½ × base × height cannot be applied directly because no height is physically measurable.
What did I learn?	Learners learned that the perpendicular height of a triangle can be expressed as h = b sin C when the hypotenuse of an internal right-angled sub-triangle is known, and that substituting this into the standard formula gives the sine-area formula A = ½ab sin C. They applied this formula to calculate the areas of Triangle A (≈ 39.9 m^2) and Triangle B (≈ 21.8 m^2) of the Kericho courtyard.

How does this explain the phenomenon?	The sine-area formula explains how the contractor can calculate the area of each triangular section of the Kericho courtyard accurately without dropping a physical perpendicular — resolving the first part of the unit puzzle. The remaining courtyard shapes (parallelogram, hexagonal border) still need their own area formulas, which will be developed in the lessons ahead.

LESSON 2 (80 minutes): Lesson 2: Triangle Area Without a Height — ½ab sin C in Practice & Heron's Formula Introduced	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to compute the area of a triangle using two sides and an included angle (½ab sin C) and to derive and apply Heron's Formula when only three sides are known, so they can solve real measurement problems in the Kericho courtyard and Rift Valley farm scenarios without needing a perpendicular height.
Knowledge	Learners will know: (a) the formula Area = ½ab sin C for a triangle given two sides and the included acute angle; (b) how to derive Heron's Formula (s = (a+b+c)/2; Area = √[s(s−a)(s−b)(s−c)]) from the sine-area formula; (c) that both formulas replace the need for a perpendicular height in real measurement contexts.
Skills	Learners will be able to: (a) measure two sides and the included angle of physical triangular objects and compute the area using ½ab sin C; (b) compare computed areas with estimates and discuss sources of error; (c) follow a structured derivation to arrive at Heron's Formula; (d) apply Heron's Formula to find the area of a triangular maize plot in the Rift Valley given only its three side lengths.
Attitudes	Learners will appreciate that mathematical formulas are tools developed to solve genuine practical problems, and will develop confidence in selecting the appropriate formula depending on the information available, mirroring the decision-making of builders, farmers, and designers in Kenya.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	In Lesson 1, learners were confronted with the Kericho courtyard puzzle and established why ½ab sin C is valid. In Lesson 2, they practise that formula with physical objects in the classroom, then encounter a new sub-phenomenon — a Rift Valley maize farmer who knows only the three side lengths of a triangular plot. This creates the need for Heron's Formula. By the end of this lesson the DQB moves from 'We can compute area if we know two sides and the included angle' to 'We can also compute area if we know only all three sides — no angle needed.' Both tools will later be applied to the Kericho courtyard triangular sections.
Safety Notes	Learners handle rulers, set-squares, and protractors — remind them not to point sharp instruments at others. If scissors are used to produce diagonal book-cover cuts, teacher should supervise and learners should cut away from the body on a stable surface.

B. LESSON OVERVIEW
Learners open by measuring two sides and the included angle of triangular physical objects (set-square faces, a diagonally-cut corner of a book cover) and computing areas using ½ab sin C, comparing results with prior estimates. The teacher then introduces a new sub-phenomenon: a Rift Valley farmer near Eldoret whose triangular maize plot has three measured side lengths but no accessible angle or height. Through a teacher-guided structured derivation (linking cos C from the cosine rule into the sine-area formula and simplifying to the semi-perimeter form), learners arrive at Heron's Formula. Small groups then apply it to the farmer's plot. The lesson closes with a DQB update and model revision: learners annotate their cumulative courtyard diagrams to show which triangular sections could use ½ab sin C and which would require Heron's Formula.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Learners receive a printed card showing a sketch of a triangular section of the Kericho courtyard labelled with two sides (e.g., 3.2 m and 4.5 m) and the included angle (52°) but NO height marked. They also see a second sketch of a triangular maize plot near Eldoret in the Rift Valley labelled with three sides (40 m, 55 m, 63 m) and the note 'No angle measured — plot is fenced on all three sides.' Learners individually write predictions: (1) Can you find the area of the courtyard triangle? How? (2) Can you find the area of the maize plot? What information would you need? Predictions are recorded in exercise books and NOT corrected yet.	Distribute the prediction cards. Say: 'Look carefully at both triangles on your card. Do not calculate yet — just think and write. You have 3 minutes of silent thinking time.' [WAIT 3 minutes — circulate and observe written responses without commenting.] After 3 minutes, cold-call 4 learners (two for each triangle): 'Amina, what did you predict for the courtyard triangle and why?' [WAIT 10 seconds after naming before accepting response.] 'Otieno, different idea?' [WAIT 10 seconds.] 'Wanjiku, for the maize plot — what information do you feel is missing?' [WAIT 12 seconds.] 'Kipchoge, do you agree or disagree with Wanjiku — explain in one sentence.' [WAIT 10 seconds.] Record the two contrasting predictions on the board under two columns: 'COURTYARD TRIANGLE' and 'MAIZE PLOT'. Do not resolve tension — tell the class: 'We will return to both of these by the end of today's lesson.'	Prediction with Contrasting Cases — by presenting two triangles that look similar but have different known information (sides+angle vs. three sides only), learners activate prior knowledge from Lesson 1 and surface the cognitive gap that motivates Heron's Formula. The contrast between the two cases makes the need for a new formula feel genuine rather than arbitrary.	Observable indicator: learner's written prediction for the courtyard triangle should reference ½ab sin C or 'use the angle' (showing Lesson 1 knowledge is retained). For the maize plot, look for learners writing 'I don't know how' or 'I need an angle or height' — this confirms the productive confusion that motivates the lesson. Teacher notes any learner who already mentions Heron's Formula or cosine rule — these learners can be assigned the extension derivation task in Phase 3.
Observe Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 [Search ARES](#)	ACTIVITY A — Hands-on Measurement (25 min): Each pair of learners receives one of three physical objects: (i) a 30-60-90 set-square (using the two shorter sides and the 30° or 60° angle), (ii) a rectangular exercise book with one corner cut diagonally creating a triangle (two sides and included angle known), or (iii) a printed scale drawing of one triangular section of the Kericho courtyard (scale 1:50). Pairs measure the two designated sides with a ruler (to the nearest mm) and the included angle with a protractor (to the nearest degree), record measurements in a table, then	Before Activity A: 'Each pair has a different object. Your job is to measure carefully — millimetres matter because a 1 mm error in a 3 m courtyard tile could mean ordering 10 extra tiles. Measure twice, record once.' Circulate during measurement; when a pair has computed their area, ask: 'How close is your computed area to your rectangle-halving estimate? What caused the difference?' [WAIT 10 seconds for response before moving on.] After all pairs have posted to the class table: 'Look at the class results. Which objects gave the largest percentage difference between	Data Collection and Comparison (Observe Phase Strategy) — learners generate their own data from physical objects, making the formula experiential rather than abstract. The class results table creates a data set that reveals measurement error, building scientific practice of analysing data. The Rift Valley sub-phenomenon is introduced as an authentic data puzzle, not a textbook exercise, sustaining the phenomenon anchor throughout the observe phase.	Observable indicators during Activity A: (1) Correct substitution — both values of a and b in metres, angle in degrees, sin computed correctly (not confused with cos). Flag any pair getting a negative area (angle entered as obtuse when it is acute — common error). (2) Percentage difference calculated as $\|computed − estimated\| / computed × 100$. During Activity B: learners' written sentences should show understanding that the missing element is an angle or a height, confirming readiness for Heron's Formula derivation.

for similar readings	compute Area = ½ab sin C using a calculator. They also estimate the area visually by enclosing the triangle in a rectangle and halving it, then compare the two values and compute the percentage difference. Results are recorded on a shared class results table drawn on the board. ACTIVITY B — New Sub-Phenomenon Introduction (5 min): Teacher projects or draws on the board a photograph description of 'Mwangi's triangular maize shamba near Eldoret — sides measured by walking and counting paces: Side 1 = 40 m, Side 2 = 55 m, Side 3 = 63 m. No angle was recorded. The plot is surrounded by a fence — dropping a perpendicular inside is not possible without destroying crops.' Learners read the scenario and write one sentence in their books: 'What maths do we need that we do not yet have?'	estimate and formula? Cold-call: Fatuma, why might the set-square give a smaller difference than the courtyard scale drawing?' [WAIT 12 seconds.] After posting the maize plot scenario: 'Write your one sentence silently — 30 seconds.' [WAIT 30 seconds.] Cold-call 3 learners: 'Atieno, what did you write?' [WAIT 10 seconds.] 'Mwangi, same or different?' [WAIT 10 seconds.] 'Kariuki, in your own words, what is the problem the farmer faces?' [WAIT 10 seconds.] Affirm: 'The farmer has exactly what surveyors in Kenya often have — three fence-line measurements and nothing else. We need a formula for that situation. That is what we build next.'		
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Structured Derivation of Heron's Formula (15 min): Learners follow a teacher-guided derivation on the board, filling in a structured notes scaffold (gaps provided on a printed or ARES-displayed template). The derivation steps are: Step 1 — Recall Area = ½ab sin C. Step 2 — From the cosine rule, cos C = $(a^2 + b^2 - c^2) / 2ab$, so $\sin^2C = 1 - \cos^2C$. Step 3 — Substitute and simplify to show $Area^2 = [s(s-a)(s-b)(s-c)]$ where s = (a+b+c)/2 is the semi-perimeter. Step 4 — Take the square root to obtain Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$. The teacher does the first two steps fully; learners complete steps 3 and 4 in pairs. PART B — Worked Example with the Maize Plot (10 min): Teacher works one full	During derivation Step 1: 'Write the sine-area formula from Lesson 1 — you have 20 seconds.' [WAIT 20 seconds.] Cold-call: 'Njeri, read your formula aloud.' [WAIT 8 seconds.] During Step 2: 'I am writing the cosine rule expression for cos C. In pairs, square both sides of $\sin^2C + \cos^2C = 1$ and substitute — you have 2 minutes.' [WAIT 2 minutes, then cold-call.] 'Omondi, what did your pair get for $\sin^2C$?' [WAIT 10 seconds.] Correct any algebraic errors immediately on the board. Before Step 3: 'This algebra gets long — I will guide you. Watch the board and fill your gaps.' Work step 3 slowly, pausing after each line: 'Does everyone see where the 2ab denominator went? Thumbs up if yes, thumbs sideways if unsure.'	Structured Derivation with Gradual Release — the teacher models the algebraically demanding steps fully, then releases the final steps to pairs, then to groups for application. This 'I do — we do — you do' sequence within the explain phase prevents cognitive overload while ensuring learners see the logical chain connecting ½ab sin C to Heron's Formula. Naming the strategy explicitly: 'We are not memorising this formula — we are understanding where it comes from, so you can always reconstruct it if you forget it in an exam.'	Observable indicators: (1) In the derivation scaffold, check that learners correctly write s = (a+b+c)/2 — a common error is s = (a+b+c). (2) In the worked example, listen for correct calculator key sequence — learners must multiply all four factors before taking the square root, not take intermediate square roots. (3) In group practice, the second Rift Valley farm (28, 35, 41): s=52, s−a=24, s−b=17, s−c=11; Area=$\sqrt{(52\times24\times17\times11)}=\sqrt{(233{,}376)}$ ≈483.1 m^2. Accept answers in range 480–486 m^2 (rounding). Flag groups whose answer differs by more than 5% — likely a calculator entry error.

	example on the board using Mwangi's farm (a=40, b=55, c=63): compute s = (40+55+63)/2 = 79; then s−a=39, s−b=24, s−c=16; Area = √(79×39×24×16) = √(1,183,104) ≈ 1,087.7 m². Teacher narrates each step aloud. PART C — Group Practice (10 min): Groups of 3–4 apply Heron's Formula to a second Rift Valley farm scenario (sides 28 m, 35 m, 41 m) and then to one triangular section of the Kericho courtyard for which only three sides are given on the scale drawing, converting scale measurements to real dimensions first.	[WAIT 5 seconds, address thumbs-sideways learners.] After Step 4: 'This formula was published by Hero of Alexandria — but Kenyan farmers have been measuring plots for centuries using rope-and-peg methods that implicitly used the same logic. The formula just makes it exact.' After the worked example: 'Before I move on — cold-call: Akinyi, what is the semi-perimeter of Mwangi's farm? Do not look at the board.' [WAIT 12 seconds.] During group practice: circulate, ask each group: 'Which formula did you choose for this courtyard section — and why this one rather than ½ab sin C?' [WAIT 10 seconds for group to discuss before answering.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one new sticky note (or fills in the ARES DQB digital card) responding to one of three prompts displayed on the board: (A) 'Something I can now calculate that I could NOT calculate in Lesson 1:'; (B) 'A question I still have about the Kericho courtyard or the Rift Valley farm:'; (C) 'Which formula would I use for which courtyard section — and why?' Notes are posted under three DQB columns: WHAT WE KNOW NOW / WHAT WE ARE STILL WONDERING / FORMULA SELECTION GUIDE. The class reads three notes aloud (one from each column, selected by the teacher). The teacher explicitly updates the driving question on the board: crossing out 'We can only find area with a height' and writing 'We can find area with (i) two sides + included angle OR (ii) three sides alone.'	Give 3 minutes of silent writing time for DQB notes. [WAIT 3 full minutes — no prompting.] Then: 'I am selecting one note from each column to read to the class.' Read Note A aloud and ask: 'Do others agree — show hands.' Read Note B aloud and say: 'That is a genuine open question — let's park it on the WONDERING column. We will answer it in Lesson 3 or 4.' Read Note C aloud and probe: 'Mwangi's note says use Heron's when you have three sides. Cold-call — Chebet, what would you need to change about a courtyard section to make ½ab sin C the better choice instead?' [WAIT 12 seconds.] Close by pointing to the updated driving question: 'The contractor in Kericho no longer needs a measuring tape dropped perpendicular across a triangular section. Two lessons in, we have two powerful tools. Three more polygon shapes remain — that is	Driving Question Board Revision — the DQB is a public, cumulative knowledge artefact. By physically striking through an old misconception ('area always needs a height') and writing the updated understanding, learners experience knowledge as growing and revisable, which mirrors authentic scientific and engineering practice. The three-column structure (Know / Wonder / Formula Guide) scaffolds metacognitive reflection.	Observable indicators: (1) Column A notes should reference ½ab sin C or Heron's Formula with correct variable labels — vague notes ('I can find area now') indicate shallow understanding; teacher marks these learners for follow-up in Lesson 3. (2) Column B notes reveal genuine misconceptions to address — common ones expected: 'What if the angle is obtuse?', 'Does Heron's work for all triangles?' — teacher photographs these for lesson planning. (3) Column C notes should correctly identify that ½ab sin C requires a known angle while Heron's requires all three sides.

		where we go next.'		
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative courtyard scale drawing begun in Lesson 1. Using the measurements from their Observe Phase Activity A and the group practice results, they: (1) Label each triangular section of the courtyard with either a '½ab sin C' tag or a 'Heron' tag, justifying their choice in a key at the bottom of the diagram. (2) Write the computed area (in m²) inside each triangular section they have solved so far. (3) Draw an arrow from the Rift Valley farm sketch (glued/drawn in the margin) to the courtyard diagram with the annotation: 'Same formula — different context.' (4) Write one sentence below the diagram: 'The contractor can now calculate ___ out of ___ triangular sections without measuring a height.' Learners who finish early annotate which sections still require formulas not yet learned (parallelogram, pentagon, hexagon) — these form the bridge to Lessons 3–5.	Distribute or ask learners to open their Lesson 1 courtyard drawings. Say: 'You have 7 minutes. Label every triangular section — formula tag, computed area if you have it, and the one sentence at the bottom.' [WAIT 7 minutes — circulate and look at diagrams.] After 7 minutes: 'Hold up your diagram. I want to see formula tags.' Scan the room quickly. Cold-call one learner whose diagram shows a correct distinction: 'Wanjiru, explain to the class why you tagged Section B as Heron's and Section C as sine-area.' [WAIT 10 seconds.] Cold-call a learner whose diagram shows a confusion: 'Kamau, I see you tagged all sections as Heron's — talk me through your thinking.' [WAIT 12 seconds — correct gently, linking back to what information is given for each section.] Close: 'Your model now has two formula tools written into it. Next lesson, we extend it to the parallelogram stage area — which has its own formula. Keep this drawing safe — it is your engineering document for the Kericho courtyard project.'	Cumulative Model Revision — the courtyard scale drawing functions as a running explanatory model that learners revise with each lesson's new formula. Each revision makes the model more complete and more powerful, mirroring how engineers and architects iteratively update drawings as they solve sub-problems. The Rift Valley → Courtyard arrow explicitly transfers learning across contexts, consolidating the understanding that formulas are transferable tools, not context-specific tricks.	Observable indicators: (1) Formula tags correctly assigned — sine-area formula where two sides and included angle are given; Heron's where only three sides are available. Misassignment is a key error to flag. (2) Computed areas written inside sections should match class results table values (within acceptable rounding). (3) The closing sentence should show a realistic count — e.g., 'The contractor can now calculate 3 out of 5 triangular sections without measuring a height.' Learners who write '5 out of 5' without having solved all sections are over-claiming — teacher notes for Lesson 3 check-in.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners distinguish correctly between the two formulas in their model-building diagrams, or did most default to one formula regardless of the given information? If more than 30% mis-tagged sections, open Lesson 3 with a 5-minute 'formula selector' retrieval activity. (2) Was the algebraic derivation of Heron's Formula too fast? If learners' scaffolded notes show gaps in Steps 3–4, consider providing a fully-worked derivation card on ARES for independent review before Lesson 3. (3) Which DQB 'Wonder' questions were most common? Questions about obtuse angles or whether Heron's works for non-Euclidean shapes signal readiness for extension; questions about basic substitution signal that more worked examples are needed before the parallelogram formula in Lesson 3. (4) Did the Rift Valley farm context feel authentic to learners? If learners seemed disengaged during the sub-phenomenon introduction, consider bringing in an actual printed land survey abstract (available from any county lands office) for Lesson 3 to strengthen the real-world anchor.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners measured two sides and the included angle of triangular classroom objects (set-squares, diagonally-cut book covers, Kericho courtyard scale drawings) and computed areas using ½ab sin C, comparing results with rectangle-halving estimates. They also read the Rift Valley maize farm sub-phenomenon: three side lengths known, no angle, no height accessible.
What did I learn?	Learners derived Heron's Formula [Area = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a+b+c)/2$] from the sine-area formula through a structured teacher-guided derivation, then applied it to Mwangi's triangular maize plot near Eldoret (sides 40 m, 55 m, 63 m → Area ≈ 1,087.7 m²) and a second practice plot.
How does this explain the phenomenon?	Learners updated their cumulative Kericho courtyard scale drawings by tagging each triangular section with the appropriate formula (½ab sin C or Heron's), recording computed areas, and writing a sentence explaining how many sections the contractor can now calculate without measuring a perpendicular height. The DQB was updated to show that area can be found with either (i) two sides and an included angle, or (ii) three sides alone.

LESSON 3 (80 minutes): Deep Practice & Model Building: Heron's Formula Consolidated	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate Heron's Formula by measuring real triangular objects in their immediate environment, computing areas, comparing results across groups, and revising their courtyard area model to include Heron's Formula as the correct tool for triangular sections where no perpendicular height is accessible.
Knowledge	Learners determine the area of a triangle given three sides using Heron's Formula in different situations (KICD SLO b).
Skills	Learners measure three sides of real triangular objects using a ruler or tape measure, compute the semi-perimeter, apply Heron's Formula step-by-step, record results in a class table, and revise a cumulative area model.
Attitudes	Learners appreciate that Heron's Formula is a powerful real-life measurement tool that resolves the limitation of needing a perpendicular height, and demonstrate unity by sharing resources and results equitably during the gallery walk.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	In Lessons 1–2, learners discovered that the courtyard contractor cannot drop a perpendicular on the triangular sections and introduced Heron's Formula as an alternative. Lesson 3 is the deep-practice consolidation: learners move from the classroom into the immediate environment, find and measure real triangles (tripod stand tops, triangular roof sections visible from the window, triangular floor tiles), apply Heron's Formula to each, and bring computed areas back to a class gallery-walk table. The lesson ends by updating the cumulative courtyard model — learners now know they can compute all triangular section areas from three side measurements alone, moving the class meaningfully closer to answering the driving question.
Safety Notes	Learners must not lean out of windows or climb structures to measure roof sections. All outdoor measurement must be done from ground level. Rulers and tape measures must be handled carefully to avoid snapping. Groups must stay within the teacher's line of sight at all times.

B. LESSON OVERVIEW

In this 80-minute lesson, learners work in groups of four to identify triangular objects in their immediate school environment (tripod stand tops, triangular tile patterns, triangular roof outlines visible from the window), measure all three sides with a ruler or tape measure, and apply Heron's Formula to compute each triangle's area. Results are recorded on a shared class gallery-walk table posted on the chalkboard. A whole-class discussion compares computed areas, surfaces errors, and consolidates procedural fluency. A 3-minute think-pair-share misconception check — "Does it matter which side you call a, b, or c in Heron's Formula?" — is followed by a cold-call debrief. The lesson closes with learners revising their cumulative Kericho courtyard area model to explicitly include Heron's Formula for the triangular sections, advancing the driving question: the contractor now has a reliable method to find the area of every triangular zone without measuring height.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner individually examines a printed scale diagram of the	Display the courtyard diagram on the board or projector. Say	Anchored Prediction (Predict-Observe-Explain cycle). By	Observable indicator: Every learner has a written numerical

📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Heron's Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Kericho courtyard (the anchoring phenomenon) showing three labelled triangular sections. The three side lengths of Triangle A are given (e.g., 4.2 m, 3.8 m, 5.1 m). Learners are asked to write, without calculating, their best estimate of Triangle A's area in square metres and to jot down one reason why their estimate might be wrong. Learners then predict: 'Will the area change if I swap which side I call a, b, or c?' They record their prediction in their exercise books before any instruction begins.	verbatim: 'Look at Triangle A in the Kericho courtyard. The contractor has measured all three sides — 4.2 metres, 3.8 metres, and 5.1 metres. In your book, estimate the area. Do not calculate yet — use your intuition. You have 2 minutes of silent thinking.' Set a visible 2-minute timer. After time, say: 'Now write this prediction: If I relabel the sides — swap which one I call a, b, or c — will the area change? Write YES, NO, or UNSURE and give one reason.' Allow 1 minute. Do NOT cold-call yet — preserve predictions for later reveal. Circulate and read 4–5 books silently to note the spread of predictions for use in Phase 3.	committing to a written estimate before calculation, learners activate prior knowledge about triangle size and surface their existing mental models — including the common misconception that labelling affects the result. These written predictions create cognitive dissonance that the lesson will resolve, making the later sensemaking more durable.	estimate and a YES/NO/UNSURE prediction with a reason in their book within 3 minutes. Teacher scans for: (1) whether learners attempt to use ½ × base × height (revealing reliance on the height formula), and (2) the distribution of YES/NO/UNSURE to calibrate the depth of the misconception check needed in Phase 3.
Observe Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Heron's Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive a Group Measurement Sheet with a table: columns for Object Name, Side a (cm), Side b (cm), Side c (cm), Semi-perimeter s, and Computed Area (cm²). Groups identify at least two triangular objects in the classroom or immediately visible from the window — for example, the top face of a tripod stand, a triangular floor tile, a triangular section of a wall display, or the triangular outline of a roof gable seen through the window. Learners measure all three sides of each triangle using a 30 cm ruler or a 3-metre tape measure. They record raw measurements, convert units if necessary, then apply Heron's Formula step-by-step: (1) compute s = (a + b + c)/2; (2) compute s−a, s−b, s−c; (3) compute s(s−a)(s−b)(s−c); (4) take the square root. Each group posts one completed row on the shared class table drawn on the chalkboard.	Before releasing groups, demonstrate the measurement protocol on a triangular cardboard cut-out: 'I place my ruler along Side 1 and read to the nearest millimetre — I get 12.4 cm. I record this as Side a. I measure Side 2 — 9.7 cm, this is Side b. Side 3 — 8.1 cm, this is Side c.' Emphasise: 'Measure to the nearest millimetre. If you use a tape measure, keep it taut against the surface.' Release groups with the instruction: 'You have 18 minutes. Measure two different triangles and complete your Group Measurement Sheet. Both group members must take at least one measurement.' Circulate every 3–4 minutes. Use targeted prompts: 'Show me where you placed the zero end of the ruler.' 'What did you get for s − b? Check it against Side b.' 'Is your answer in centimetres or metres — be consistent.' After 18 minutes, call groups back: 'One representative from each group, write your best triangle row on the class table on	Concrete-to-Abstract Bridging through Real-Object Measurement. By measuring actual objects in their environment rather than working from textbook diagrams, learners tie the abstract formula to physical reality — directly mirroring the contractor's situation in the Kericho courtyard. The shared class table transforms individual measurements into a collective evidence base, making patterns (and errors) visible to all.	Observable indicators: (1) Each group's Measurement Sheet has both triangles fully filled in with consistent units. (2) The class table on the board has at least one row per group. Teacher checks three specific things while circulating: (a) that learners measure actual side lengths, not estimates; (b) that the semi-perimeter s is correctly computed as (a+b+c)÷2, not (a+b+c); (c) that the square root step is applied to the full product s(s−a)(s−b)(s−c), not just s. Flag any group that squares instead of square-roots for immediate intervention.

		the board. Write the object name, the three sides, s, and your area.' Allow 5 minutes for posting. Instruct remaining learners: 'While your representative writes, check your own calculation one more time.'		
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Heron's Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole class conducts a gallery walk of the class table on the board. Learners examine all rows, identify the largest and smallest areas, and check whether any two groups measured the same object and got different answers (measurement error discussion). The teacher then launches the misconception check: 'Does it matter which side you call a, b, or c in Heron's Formula?' Learners do a 3-minute think-pair-share — first 1 minute silent individual thinking, then 2 minutes discussing with their partner — before the cold-call debrief. Pairs are asked to construct a proof-by-example: take one of the triangles from the class table and re-compute the area swapping a and b. They compare the two results. Learners then write a one-sentence generalisation in their books: 'The area computed by Heron's Formula does / does not change when the labels a, b, c are swapped, because ___.' The teacher connects this back to the Kericho courtyard: 'This means the contractor can label the sides of each triangular section in any convenient order and still get the correct area — this is very useful on a real construction site.'	Begin the gallery walk by saying: 'Walk to the board, spend 3 minutes reading all the rows. Do not talk yet.' After 3 minutes: 'Back to your seats.' Then cold-call: point to a specific row — 'Group 3 measured the tripod stand and got an area of [X] cm². Group 5 also measured the tripod stand and got [Y] cm². The difference is [Z] cm². What could explain this?' Wait 12 seconds. Cold-call one learner by name: 'Akinyi, what do you think?' After response, cold-call a second learner: 'Mwangi, do you agree or do you have a different idea?' Validate measurement error as a real engineering concern. Then launch the misconception check: 'Now I want you to think about this question alone — does it matter which side you call a, b, or c? One minute, silent. Write your thinking.' After 1 minute: 'Turn to your partner, 2 minutes, share and challenge each other.' After 2 minutes, cold-call three pairs in sequence: 'Otieno, what did your pair decide — and what was your proof-by-example?' After each response, press: 'Can someone add to that or challenge it?' After the third cold-call, provide explicit closure: 'Heron's Formula uses a product — multiplication is commutative. $s(s-a)(s-b)(s-c)$ gives the same result regardless of which side is labelled a, b, or c. The formula is symmetric. Write that word — symmetric — in your	Think-Pair-Share with Proof-by-Example followed by Explicit Generalisation. This two-step strategy — first constructing a numerical counterexample to test the conjecture, then articulating a general principle — develops mathematical reasoning beyond procedural recall. The explicit vocabulary instruction ('symmetric') gives learners a precise word to anchor the concept. The immediate connection to the courtyard phenomenon ensures the explanation phase remains anchored to the unit's driving question.	Observable indicators: (1) Every learner has a completed one-sentence generalisation in their book using the word 'symmetric' or equivalent reasoning. (2) During cold-call, at least two of the three pairs can articulate not just the correct answer ('No, it does not matter') but a reason ('because multiplication is commutative / the formula is symmetric'). If fewer than half the class writes a correct generalisation, the teacher pauses and works a second proof-by-example on the board before proceeding.

		book.' Finally, connect explicitly to the phenomenon: 'For the Kericho contractor, this means he can measure the sides of each triangular section in whatever order is physically convenient on site and the calculation is still valid.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Heron's Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a sticky note (or writes in a designated DQB section of their book). They respond to two prompts: (1) 'What new evidence from today's lesson helps me answer the driving question: How can we calculate the exact area of any polygon-shaped surface even when we cannot measure its height?' (2) 'What question do I still have about the Kericho courtyard?' Learners post sticky notes on the class DQB under two columns: WHAT WE NOW KNOW and WHAT WE STILL WONDER. The teacher reads four notes aloud (two from each column) and explicitly links today's evidence to the driving question: 'We now have a method — Heron's Formula — that lets us compute the area of any triangle from three side measurements alone. We no longer need height. This directly solves the contractor's problem for every triangular section of the Kericho courtyard.'	Distribute sticky notes or direct learners to their DQB book section. Say: 'Two minutes, silent writing. Respond to both prompts.' After 2 minutes, circulate and read notes as learners post them. Select and read aloud two KNOW notes and two WONDER notes — choose ones that are specific and directly connected to the courtyard. After reading each, ask the class: 'Does anyone want to add to this?' Wait 8 seconds before moving on. Close with: 'Look at the WONDER column. We will not answer all of these today — some will be answered in Lessons 4, 5, and 6. Hold onto your questions. They are good ones.' Point to any note about hexagonal tiles or the parallelogram stage and say: 'This question — about the hexagonal border tiles — is exactly what Lesson 5 will address.'	Driving Question Board as a Cumulative Evidence Tracker. The DQB makes learners' growing understanding visible and public, reinforcing that scientific and mathematical learning is incremental. By explicitly naming what is now resolved (triangular sections via Heron's Formula) and what remains open (hexagonal tiles, parallelogram stage), learners develop metacognitive awareness of the unit storyline and feel the forward momentum of inquiry.	Observable indicator: Every learner has posted at least one sticky note under WHAT WE NOW KNOW that references Heron's Formula and at least one under WHAT WE STILL WONDER that is specific to the courtyard (not generic). Teacher flags any KNOW notes that misstate the formula or confuse it with the sine-area formula from Lesson 1 — these learners are marked for follow-up in the next lesson's opening review.
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners retrieve the cumulative Kericho courtyard area model they began in Lesson 1 (a labelled scale sketch of the courtyard showing the triangular sections, parallelogram stage, and hexagonal border). Working individually, learners revise their model by: (1) Writing the Heron's Formula annotation beside each triangular section: 'Area = √[s(s−a)	Say: 'Take out your courtyard model from Lesson 1. You have 10 minutes to revise it based on everything we have learned today and in Lessons 1 and 2.' Write the three revision tasks on the board: '1. Annotate each triangle with Heron's Formula and one computed area. 2. Shade triangular sections and write SOLVED. 3. Mark remaining	Cumulative Model Revision as a Sense-of-Progress Mechanism. Revisiting and annotating the same courtyard diagram across lessons creates a tangible artefact of growing understanding. The SOLVED / PENDING coding scheme makes the unit's storyline structure visible — learners can literally see how much of the driving question they have	Observable indicators: (1) Every learner's model has Heron's Formula correctly written beside at least one triangular section, with a numerical area computed using the courtyard's given dimensions. (2) Learners have correctly identified the parallelogram and hexagonal zones as PENDING. (3) During the partner share, the teacher listens for whether

📖 **HTML:** [Heron's Formula (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(s−b)(s−c)] — use when three sides are known, no height needed'; (2) Recording their computed area estimates for at least one triangular section using the side lengths given on the scale diagram; (3) Shading the triangular sections in one colour and writing 'SOLVED ✓' to indicate that a reliable calculation method now exists for these zones; (4) Identifying the zones still unresolved (parallelogram stage, hexagonal border) and marking them 'PENDING — Lessons ahead.' Learners share their revised model briefly with a partner and note one difference between their two models.	zones PENDING.' Circulate and check that annotations are mathematically correct. For learners who finish early, prompt: 'Can you write a sentence the contractor could use to explain to his client why Heron's Formula is reliable even without height measurements?' With 2 minutes remaining, instruct: 'Share your model with your partner. Identify one thing that is different between your two models — not wrong, just different.' Cold-call one pair: 'Kamau and Wangari — what was one difference you noticed?' After their response, close with: 'Your model is growing lesson by lesson. By Lesson 8, every zone of the courtyard will be labelled SOLVED. Today we completed the triangular sections. Well done.'	answered and how much remains, sustaining motivation and cognitive engagement across eight lessons.	learners can articulate why Heron's Formula removes the need for height — if a learner cannot explain this, the teacher prompts: 'What information did you need for the old base × height formula that you do NOT need for Heron's Formula?' Exit check: Teacher collects three randomly selected models at the end of the lesson to review annotations for accuracy before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the gallery walk, were measurement errors large enough to prompt a productive discussion about precision — or were errors negligible? If negligible, consider introducing deliberate 'noisy' measurements in future lessons to deepen the error-analysis conversation. (2) During the think-pair-share misconception check, how many learners initially answered YES (the label matters)? If more than one-third answered YES, plan a short 5-minute revisit at the start of Lesson 4 using a second proof-by-example with different numbers. (3) Were all groups able to locate and measure two distinct triangular objects in the classroom environment? If some groups struggled to find triangles, prepare a backup set of printed triangular cut-outs with pre-marked vertices for future lessons. (4) Are learners' cumulative courtyard models accurate and growing in detail? Review the three collected models tonight — if any show Heron's Formula mis-stated (e.g., missing the square root, or using perimeter instead of semi-perimeter), address these specific errors explicitly in Lesson 4's opening. (5) Did the DQB WONDER column surface any questions about the hexagonal tiles or parallelogram stage that could be used as a compelling hook for Lesson 4? Photograph the DQB and use the most specific WONDER notes as the opening provocation for Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners measured three sides of real triangular objects in the classroom environment (tripod stand tops, triangular tiles, roof outlines visible from windows) using rulers and tape measures, and recorded all measurements on a Group Measurement Sheet. They posted results on a shared class table on the chalkboard and conducted a gallery walk comparing computed areas across groups.
What did I learn?	Learners consolidated Heron's Formula — Area = $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$ — and proved through proof-by-example that the formula is symmetric: the computed area does not change when the labels a, b, and c are swapped. They

	connected this to the Kericho courtyard context: the contractor can measure the three sides of each triangular section in any convenient order and still obtain the correct area without needing a perpendicular height.
How does this explain the phenomenon?	Learners revised their cumulative Kericho courtyard area models by annotating each triangular section with Heron's Formula, recording a computed area estimate, shading solved zones, and marking the parallelogram stage and hexagonal border as PENDING for future lessons. They updated the Driving Question Board with new evidence under WHAT WE NOW KNOW and posted specific questions about remaining unresolved zones under WHAT WE STILL WONDER.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Area of Rhombus and Parallelogram — Measuring the Stage Floor Without Dropping a Perpendicular

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners develop and apply the formulas $A = \frac{1}{2}d_1d_2$ (rhombus) and $A = ab \sin \theta$ (parallelogram) so that the Kericho courtyard contractor can calculate the area of the parallelogram-shaped stage floor and rhombus-shaped fabric tile patches without needing a measurable perpendicular height.
Knowledge	Learners will know: (i) the diagonal property of a rhombus and why $A = \frac{1}{2}d_1d_2$ works; (ii) how the sine of an included angle replaces the need for a perpendicular height in $A = ab \sin \theta$ for a parallelogram; (iii) the structural link between $A = \frac{1}{2}ab \sin C$ (triangle, Lesson 1) and $A = ab \sin \theta$ (parallelogram — two congruent triangles).
Skills	Learners will be able to: (i) measure diagonals of a rhombus-shaped object and compute its area; (ii) measure two adjacent sides and the included angle of a parallelogram-shaped object and compute its area using $A = ab \sin \theta$; (iii) connect the parallelogram formula to the triangle formula from Lesson 1; (iv) update the Driving Question Board with new evidence.
Attitudes	Learners appreciate that mathematics provides tools for solving real measurement problems faced by builders and textile designers in their community; they demonstrate responsibility by handling measuring instruments carefully and unity by sharing resources during group work.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	In Lesson 1 learners discovered $A = \frac{1}{2}ab \sin C$ for a triangle when no perpendicular height was measurable. Lesson 4 advances the storyline: the Kericho courtyard contractor now faces the parallelogram-shaped stage area and rhombus-shaped decorative fabric patches. Learners realise that a parallelogram is simply two congruent triangles joined together — so the sine rule they already trust doubles to give $A = ab \sin \theta$. The rhombus diagonal formula emerges from the same perpendicular-bisector property learners can see physically. Both results update the DQB and move learners one step closer to pricing the entire courtyard.
Safety Notes	Learners should handle rulers and geometrical set instruments (compasses, protractors) with care — pointed ends must face downward when not in use. Scissors used to cut card offcuts must be handled responsibly and passed handle-first. Ensure adequate desk space so that measuring does not result in instruments falling.

B. LESSON OVERVIEW

Learners bring or identify objects with rhombus and parallelogram shapes — kanga fabric patches, parallelogram-shaped card offcuts, and floor-tile arrangements from photographs of the Kericho courtyard scale drawing. Working in groups of four, they first measure the two diagonals of a rhombus-shaped patch and derive $A = \frac{1}{2}d_1d_2$ by decomposing the rhombus into four right triangles. They then measure two adjacent sides and the included angle of a parallelogram-shaped card offcut, apply $A = ab \sin \theta$, and — guided by the teacher — explicitly connect this formula to the triangle formula $A = \frac{1}{2}ab \sin C$ from Lesson 1 (the parallelogram = 2 triangles bridge). The teacher uses cold-calling, targeted questioning, and a brief class discussion to consolidate both formulas before learners apply them to the courtyard stage area and update the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually examines two printed cards: Card A shows the parallelogram-shaped stage area from the Kericho courtyard scale drawing (two side lengths and included angle labelled: a = 6 m, b = 4 m, θ = 65°, no height given). Card B shows a rhombus-shaped kanga fabric patch (diagonals d_1 = 8 cm, d_2 = 5 cm labelled, no height given). Learners write in their exercise books: (i) a prediction of which shape has the larger area; (ii) a method they would try to calculate each area — even if they are unsure. Learners then share predictions with a shoulder partner for 2 minutes.	Distribute Card A and Card B before learners sit. Say: 'Before we measure anything, I want you to THINK ALONE for three minutes. Write your prediction and your proposed method in your book — even a wrong method is useful today.' [Allow 3 minutes silent thinking — WAIT TIME enforced.] Then say: 'Share your prediction with the person next to you. You have 2 minutes.' Circulate and note 2–3 interesting or misconception-laden predictions. After sharing, cold-call 3 learners (one who predicted triangle-based decomposition, one who mentioned base × height, one who was uncertain): 'Aisha, what method did you suggest for the parallelogram?' [WAIT 10 seconds before prompting.] Record the three methods on the board under the heading: 'Our Predictions — Lesson 4'. Do NOT confirm or correct yet. Say: 'Keep these predictions visible — we will return to them at the end of the lesson.'	Predict-Before-Observe (Anticipatory Prediction): By articulating a method before instruction, learners activate prior knowledge (base × height from earlier grades, ½ab sin C from Lesson 1) and create a cognitive gap when the standard method fails — motivating genuine inquiry into the sine-based formula.	Observable indicator: Every learner has written at least one proposed method in their book before partner sharing begins. Teacher scans books during circulation — look specifically for learners who write 'base × height' (common prior method) vs those who write '½ab sin C × 2' (Lesson 1 transfer). These two groups will be strategically cold-called in Phase 3.
Observe Phase 📹 **VIDEO:** Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook)	ACTIVITY 1 — Rhombus diagonal formula (12 minutes): Groups of four receive a rhombus-shaped card offcut (pre-cut to approximately 10 cm × 10 cm) and a ruler. Step 1: Learners draw both diagonals and label their measured lengths d_1 and d_2. Step 2: Learners cut (or fold) the rhombus along one diagonal, then cut each half along the second diagonal — producing four right triangles. Step 3: Each group records in a table: the base and height of one right triangle, calculates its area, multiplies by 4, and compares to $½d_1d_2$. They	Before Activity 1, hold up a pre-made rhombus card and say: 'The contractor in Kericho has rhombus-shaped kanga fabric patches for the border decoration. He knows only the diagonal lengths — no height. Your job in the next 12 minutes is to discover a formula he can use.' [Distribute materials.] Circulate and prompt struggling groups: 'What angle do the diagonals make where they cross? Measure it.' [WAIT 10 seconds for response.] After 12 minutes, pause the class: cold-call one group spokesperson to share findings. Say: 'Fatuma, your group	Decomposition-and-Recomposition (Concrete Manipulation): Physically cutting the rhombus into four right triangles and the parallelogram into two congruent triangles makes the abstract formulas emerge from visible, tangible structure. Learners see WHY $½d_1d_2$ works (four right triangles with legs $d_1/2$ and $d_2/2$) and WHY A = ab sin θ = 2 × ½ab sin θ (two identical triangles). This anchors the formula in geometric reasoning rather than rote memory.	Observable indicators: (1) In the rhombus table, the column 'area of 4 triangles' and the column '$½d_1d_2$' should agree to within measurement error (±0.5 cm^2) — teacher checks at least 3 group tables during circulation. (2) On the class poster, every group's parallelogram area formula must show the step 'A = 2 × ½ab sin θ = ab sin θ' — not just the final answer. Flag any group that writes only the numerical answer without the derivation step for targeted questioning in Phase 3.

Source: CK-12 🔍 Search ARES for similar readings	complete the sentence: 'The area of a rhombus equals ½ × ___ × ___ because the diagonals bisect each other at ___°.' ACTIVITY 2 — Parallelogram sine formula (13 minutes): Each group receives a parallelogram-shaped card offcut. Step 1: Measure sides a and b with a ruler, and the included angle θ with a protractor. Step 2: Using scissors, cut the parallelogram along one diagonal — producing two triangles. Step 3: Using the formula from Lesson 1 (A = ½ab sin C), calculate the area of one triangle. Step 4: State the total area of the parallelogram. Step 5: Groups write the general formula A = ab sin θ and record their numerical result on a shared class results poster on the board.	found what formula?' [WAIT 10 seconds.] Write $A = \frac{1}{2}d_1d_2$ on the board. Before Activity 2, point to the Lesson 1 formula on the DQB and say: 'Look at what we posted in Lesson 1 — A = ½ab sin C. Today I want you to cut your parallelogram into triangles. What do you notice?' [WAIT 15 seconds.] After Activity 2, cold-call 2 groups (total 2 cold-calls) to read their formula and numerical result from the poster. Say: 'Every group, check that your result is on the poster before we move to explaining.'		
Explain Phase 📹 **VIDEO:** Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	WHOLE-CLASS CONSOLIDATION (10 minutes): The teacher leads a structured discussion connecting A = ab sin θ back to A = ½ab sin C (Lesson 1) and to the Kericho courtyard context. Learners then work individually on two applied problems in their exercise books: Problem 1 (Rhombus): A rhombus-shaped decorative tile on the Kericho courtyard border has diagonals of 12 cm and 9 cm. Calculate its area. Problem 2 (Parallelogram stage): The parallelogram stage area has adjacent sides of 6 m and 4 m with an included angle of 65°. Calculate the area to 2 decimal places (use sin 65° = 0.9063). Learners write their full working and state the answer in appropriate units. EXTENSION (for fast finishers): A kanga fabric design uses a parallelogram with sides 15 cm and 10 cm. The designer knows	Open the consolidation by pointing to both formulas on the board and saying: 'In Lesson 1 we found A = ½ab sin C for a triangle. Today we found A = ab sin θ for a parallelogram. Hassan — stand up — can you explain in ONE sentence why the parallelogram formula is exactly double the triangle formula?' [WAIT 15 seconds. If no response, prompt: 'How many triangles did your group cut from the parallelogram?'] Cold-call 1 more learner to confirm. Then say: 'So the contractor in Kericho can calculate the stage area WITHOUT measuring a height. Write this in your own words as a sentence under today's date.' [Allow 2 minutes writing.] Move to individual problems. Say: 'You have 8 minutes — work in silence. Show every step. I will be looking at whether you write the formula first before substituting numbers.'	Formula Bridging (Explicit Connection): The teacher explicitly names and draws an arrow on the board linking A = ½ab sin C → A = ab sin θ, labelling it '×2 because 2 triangles = 1 parallelogram'. This bridging strategy prevents learners from storing two disconnected formulas and instead builds a coherent mental model of how polygon area formulas are related — directly supporting the storyline goal of building a versatile toolkit for the courtyard problem.	Observable indicators: (1) In individual work, learners write the formula before substituting (not just the answer) — teacher collects 5 random books after the phase and checks for this. (2) Problem 2 correct answer: A = 6 × 4 × sin 65° = 24 × 0.9063 ≈ 21.75 m² — any answer in the range 21.70–21.80 m² (due to rounding) is accepted. (3) During cold-calling, the nominated learner can state in their own words why the parallelogram formula is double the triangle formula — this is the key conceptual indicator for Lesson 4.

	only that sin θ = 0.75. Calculate the area and explain whether a 200 cm² piece of backing fabric is sufficient.	Circulate; look for: unit errors (cm vs m), forgetting to multiply by sin θ, calculator errors with sin 65°. After 8 minutes, cold-call one learner for Problem 1, one for Problem 2 (2 cold-calls total). Write the worked solutions on the board, narrating each step aloud. Ask the class: 'Does anyone have a different answer? Raise your hand.' Address discrepancies immediately.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four revisit the class DQB (a large poster or section of the board maintained since Lesson 1). Each group receives two sticky notes of different colours: ORANGE = 'What we can now calculate about the Kericho courtyard'; GREEN = 'What we still cannot calculate or are unsure about'. Groups discuss for 3 minutes and write one orange and one green note. One spokesperson from each group walks up and places both notes on the DQB, reading the orange note aloud. The class also physically draws a tick (✓) next to the DQB question posted in earlier lessons: 'Can we find the area of shapes that have no right angle and no measurable height?' — because today's formulas answer this for rhombuses and parallelograms.	Say: 'It is time to update our Driving Question Board. Remember — the contractor in Kericho is waiting for our calculations. In groups, discuss: what can we NOW calculate that we could not calculate before Lesson 4? And what is still unsolved?' [Allow 3 minutes group discussion.] Distribute sticky notes. After groups post their notes, point to the DQB and say: 'Look at the question we wrote in Lesson 1 about shapes with no measurable height. James — does today's lesson answer that question for the stage area? Tell me yes or no and why.' [WAIT 10 seconds.] Draw the tick on the DQB together with the class. Then say: 'What questions remain? We will need those for Lessons 5 and 6.' Read 2 green sticky notes aloud, leaving them visible on the DQB as anchors for upcoming lessons. Add today's two formulas — $A = \frac{1}{2}d_1d_2$ and $A = ab \sin \theta$ — to the 'Formula Toolkit' section of the DQB.	Driving Question Board (DQB) as a Living Knowledge Artefact: Publicly marking which courtyard sub-problems are now solvable (orange notes) versus still open (green notes) externalises the class's growing understanding and maintains narrative momentum in the storyline. Learners can see their own intellectual progress across the 8-lesson sequence, which builds self-efficacy and mathematical confidence.	Observable indicators: (1) Every group's orange sticky note references a specific part of the Kericho courtyard (stage area or fabric patches) — not a generic statement. (2) The class correctly identifies and ticks the DQB question about no measurable height. (3) At least 2 green sticky notes raise questions about pentagon or hexagon area (previewing Lessons 5–6) or about what happens when neither diagonal nor angle is given — demonstrating that learners are thinking ahead within the storyline.
Model Building Phase 📹 **VIDEO:** Proof: Rhombus diagonals are	Each learner updates their personal cumulative courtyard model — a scale drawing begun in Lesson 1 showing the Kericho courtyard floor plan. Today's task: (1) Shade and label the	Say: 'Open your courtyard model from Lesson 1. You have 10 minutes to update it with today's new information. Your model must show: the shaded stage area with its calculated value, one rhombus	Cumulative Model Revision (Running Scientific Model): Requiring learners to update the same scale-drawing model across all 8 lessons creates a tangible record of knowledge growth. The	Observable indicators: (1) The stage area on every learner's model shows a numerical value between 21.70 and 21.80 m² with correct units (m²). (2) The Formula Toolkit box on every model

[perpendicular bisectors](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	parallelogram stage area on the scale drawing and write the calculated area (≈ 21.75 m²) inside it. (2) Shade and label one rhombus fabric patch and write the formula $A = \frac{1}{2}d_1d_2$ next to it. (3) In the 'Formula Toolkit' box at the bottom of the model sheet, add: $A\_rhombus = \frac{1}{2}d_1d_2$ and $A\_parallelogram = ab \sin \theta$, with a note: 'Related to $A\_triangle = \frac{1}{2}ab \sin C$ (Lesson 1) — parallelogram = 2 triangles.' (4) Annotate with one sentence: 'The contractor can now calculate the stage area and fabric patch areas without measuring a height.' Groups then briefly compare their updated models to check for consistency in shading and labelling.	patch with its formula, and a note connecting today's formulas to Lesson 1.' [Circulate and check that learners are writing the formula connection note — prompt those who only write the numbers: 'Where does the ab sin θ come from? Connect it to what we wrote in Lesson 1.'] [WAIT 10 seconds before prompting.] After 10 minutes, say: 'Hold up your model. Look at your neighbour's model. Do your stage area calculations match? If not, discuss why — you have 2 minutes.' Cold-call 1 pair who had a discrepancy to share what caused it (likely unit confusion or rounding). Close by saying: 'Your model is your personal contractor's toolkit. By Lesson 8 it will contain every formula needed to order all the materials for the Kericho courtyard. Today we added two more tools.'	annotation connecting $A = ab \sin \theta$ to $A = \frac{1}{2}ab \sin C$ (Lesson 1) enforces the cross-lesson conceptual link and prevents compartmentalisation of formulas. The model also serves as a revision artefact for summative assessment.	contains BOTH new formulas with correct notation. (3) The annotation sentence explicitly mentions 'no height needed' or 'sine replaces height' — this is the exit-ticket-level indicator that the lesson's core concept has been internalised. Teacher collects 6 models (2 per ability level) at the end of the lesson as a formative sample.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) FORMULA BRIDGING — During Phase 3, could learners articulate in their own words why $A = ab \sin \theta$ is exactly double $A = \frac{1}{2}ab \sin C$? If fewer than two-thirds of cold-called learners could do so, plan a brief warm-up in Lesson 5 that revisits the two-triangle decomposition using a different parallelogram. (2) MEASUREMENT ACCURACY — In Phase 2, did groups' calculated areas from four right triangles agree with $\frac{1}{2}d_1d_2$ to within acceptable error? If not, was the issue ruler technique, protractor reading, or cutting precision? Consider whether a digital GeoGebra activity (ARES offline simulation) would provide cleaner data in a future iteration. (3) DQB QUALITY — Were learners' green sticky notes genuinely predictive of Lessons 5–6 content (pentagon, hexagon), or were they vague? If vague, spend 2 extra minutes in the next DQB update modelling what a specific, storyline-anchored question looks like. (4) COURTYARD MODEL — Did learners connect the formula note on their model back to Lesson 1 without prompting? Self-initiated connection is the highest indicator of conceptual transfer — note which learners did this for differentiated extension in Lessons 5–6. (5) INCLUSION — Were learners who struggled with sine (from Lesson 3 Trigonometry sub-strand) able to participate meaningfully? If not, consider pairing them with a peer who can demonstrate sin 65° on a calculator before the individual work phase in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically cut rhombus-shaped card offcuts into four right triangles and parallelogram-shaped card offcuts into two congruent triangles, measured diagonals and side lengths, and recorded results on a class poster — demonstrating that both shapes can be decomposed into triangles whose dimensions are directly measurable.

What did I learn?	Learners derived and applied $A = \frac{1}{2}d_1d_2$ for the rhombus (from the perpendicular-bisecting diagonal property) and $A = ab \sin \theta$ for the parallelogram (as twice the area of one constituent triangle), and explicitly connected the parallelogram formula to the triangle formula $A = \frac{1}{2}ab \sin C$ established in Lesson 1.
How does this explain the phenomenon?	Learners explained — in writing on their courtyard models and verbally during cold-calling — that the Kericho courtyard contractor can calculate the area of the parallelogram stage floor ($\approx 21.75 \text{ m}^2$) and rhombus fabric patches using only side lengths, diagonal lengths, and an included angle, without needing to physically measure a perpendicular height, because the sine of the included angle mathematically encodes the height.

LESSON 5 (80 minutes): Practise & Apply — Rhombus and Parallelogram in Context	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate and apply both parallelogram/rhombus area formulas — the diagonal method and the sine method — to solve structured real-life problems set in the Kericho courtyard scenario and related Kenyan contexts, and decide which formula is more practical given available measurements.
Knowledge	Learners know that: (i) the area of a parallelogram = base × height = ab sin θ, where a and b are adjacent sides and θ is the included angle; (ii) the area of a rhombus = ½ d_1d_2, where d_1 and d_2 are the diagonals; (iii) when perpendicular height is unavailable but two sides and an included angle are known, the sine formula is the practical choice; (iv) when both diagonals are measurable, the diagonal formula is efficient.
Skills	Learners can: (i) identify which measurements are available in a given problem and select the appropriate formula; (ii) accurately compute areas of parallelograms and rhombuses using both methods; (iii) compare answers obtained by both methods where both sets of data are given; (iv) use a peer-assessment checklist to evaluate a classmate's solution; (v) annotate and update the cumulative Kericho courtyard scale-drawing model with the calculated parallelogram stage area.
Attitudes	Learners appreciate that choosing the right mathematical tool depends on what can realistically be measured in a field situation, and that checking work with an alternative method builds confidence and accuracy — qualities valued by builders, farmers, and designers in their communities.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	In Lessons 3 and 4 learners derived and first applied the parallelogram/rhombus formulas. Lesson 5 is the Practise & Apply phase: learners now work independently and in pairs on a structured problem set drawn directly from the Kericho courtyard (the parallelogram stage area), a kanga cloth patch, and a school garden — cementing fluency, exercising formula-selection judgement, and updating the shared courtyard model so the class can see the cumulative progress toward answering the driving question.
Safety Notes	No sharp instruments or chemicals are used. Learners working on floor-level model sheets should be reminded not to kneel on each other's work. Ensure adequate lighting for reading scale drawings.

B. LESSON OVERVIEW
Learners open with a rapid predict-then-check warm-up that surfaces any residual confusion about when to use the sine method versus the diagonal method. They then work through a three-part structured problem set — (A) the Kericho courtyard parallelogram stage, (B) a kanga cloth patch problem, (C) a school garden in Kisumu — each problem supplying different combinations of measurements so learners must consciously select their formula. A structured peer-assessment checklist is used after Part B, giving learners metacognitive language for evaluating mathematical reasoning. The lesson closes with learners physically annotating the class courtyard model with the stage area value, advancing the driving question board, and a teacher-led whole-class debrief that surfaces the core idea: the sine formula is the field-practical tool when heights cannot be measured.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-sheet showing two parallelogram diagrams side by side. Diagram A shows the Kericho courtyard stage: sides 8 m and 5 m, included angle 65°, but no height marked. Diagram B shows the same stage redrawn with a perpendicular height of 4.53 m added (no angle given). Learners individually write: (i) which formula they would use for each diagram and why, (ii) a numerical estimate of the area for Diagram A without calculating — just a 'sense-check' range (e.g. 'between 30 m² and 45 m²'). They do NOT yet compute.	Distribute the half-sheets face-down. Say: 'Before you turn it over, think back to our last two lessons — what two methods do we have for finding the area of a parallelogram?' Wait 10 seconds, then cold-call 3 learners. After responses, say: 'Now turn over. You have 4 minutes — write your formula choice and your sense-check range. No calculators yet.' Circulate silently; scan for learners who write 'base × height' for Diagram A (common error — they cannot compute it without the height). After 4 minutes, cold-call 2 learners to share their Diagram A choice. Probe: 'What stops you using base × height for Diagram A?' Wait 12 seconds. Do NOT confirm or correct yet — record the class split on the board as a tally (sine method vs. base×height vs. unsure).	Activating Prior Knowledge & Surfacing Misconceptions — the side-by-side diagrams force learners to notice that the same shape can be described with different measurement sets, making the formula-selection decision salient before any computation begins. The tally on the board creates a public, revisable record that will be resolved by the end of the phase.	Observable indicator: learner writes 'sine formula / ab sin θ' for Diagram A and gives a plausible sense-check range (20–50 m²). Red flag: learner writes 'base × height' for Diagram A — teacher notes these learners for targeted support in Phase 3.
Observe Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners now work on the three-part structured problem set (printed card or displayed on screen). PART A — Kericho Courtyard Stage (parallelogram): The contractor's notes show side a = 8 m, side b = 5 m, included angle θ = 65°. No height is accessible. Task: Compute the area using the sine formula. Show all steps. PART B — Kanga Cloth Patch (rhombus): A tailor in Nairobi's Gikomba market has a rhombus-shaped decorative kanga patch. The two diagonals measure 24 cm and 18 cm. Task: Compute the area using the diagonal formula. Then the tailor later discovers the included angle between two adjacent sides is 56° and each side is 15.23 cm. Task:	Read Part A aloud together: 'The contractor standing on the courtyard stage in Kericho has only his tape measure and a protractor — he cannot drop a plumb line to find the height. Which formula does he reach for?' Wait 10 seconds. Then say: 'Begin Part A. Show every step — formula, substitution, calculation, answer with units.' Circulate during individual work. Stop at any learner who is multiplying 8 × 5 directly and ask quietly: 'What does that formula assume about the angle?' For Part B, after learners finish both calculations, say: 'Your two answers for Part B should be very close. If they differ by more than 1 cm², hunt for your arithmetic error before moving on.'	Structured Problem Sequencing — Parts A, B, and C are deliberately ordered: A requires only the sine formula (no diagonals given); B requires both formulas on the same shape so learners can verify consistency; C requires the sine formula but with an obtuse angle (110°), which tests whether learners know sin(110°) = sin(70°) and can handle this without panic. This scaffolded sequence builds procedural fluency while continuously connecting back to the Kericho phenomenon.	Observable indicators: (i) Part A — correct setup: Area = 8 × 5 × sin 65° ≈ 36.25 m²; (ii) Part B — diagonal answer: ½ × 24 × 18 = 216 cm²; sine answer: 15.23² × sin 56° ≈ 215.9 cm² (within rounding); (iii) Part C — Area = 6.5 × 4 × sin 110° ≈ 24.44 m²; explanation mentions non-right angle. Red flags: learner uses sin(110°) as negative (calculator mode error) — address immediately by demonstrating degree mode check.

	Compute the area again using the sine formula and compare the two answers. PART C — School Garden, Kisumu (parallelogram): A school in Kisumu wants to tile a parallelogram-shaped garden bed with stone slabs. Side lengths are 6.5 m and 4 m; the included angle is 110°. Task: Compute the area; then explain in one sentence why the contractor cannot simply multiply 6.5 × 4. Learners work individually for 18 minutes, then compare answers with a shoulder partner for 4 minutes.	For Part C, after shoulder-partner comparison, cold-call 1 pair: 'Walk us through your Part C explanation sentence.' Write key words from their answer on the board. Ensure the phrase 'the angle is not 90°' appears before moving on.		
Explain Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners use a structured peer-assessment checklist (provided on a small card) to assess their shoulder partner's Part B solution. The checklist has five items: (1) Correct formula stated for diagonal method ✓/✗; (2) Correct substitution of d_1 and d_2 ✓/✗; (3) Correct formula stated for sine method ✓/✗; (4) Correct substitution of side and angle ✓/✗; (5) Comparison sentence present and mathematically valid ✓/✗. After 6 minutes of peer assessment, the class reconvenes for a teacher-facilitated whole-group discussion structured around three questions: (Q1) 'For the Kericho stage, which formula was the contractor forced to use — and why?' (Q2) 'For the Nairobi kanga patch, both formulas worked. When would you prefer the diagonal method in a real workshop or market?' (Q3) 'For Part C, did anyone get a different area when they tried base × height? What went wrong?' The teacher then formalises the decision rule on the board: USE DIAGONAL METHOD when both diagonals are physically	Distribute peer-assessment checklist cards. Say: 'You are not marking your partner right or wrong — you are being a thinking partner. For each item, if it is wrong, write a hint, not the answer.' After 6 minutes, bring class together. Post Q1 on board; wait 12 seconds; cold-call 2 learners (choose one who predicted 'base×height' in Phase 1 to reveal their revision). For Q2, say: 'Think about a tailor at Gikomba measuring a kanga on a cutting table. What is easier — measuring the diagonals with a tape, or measuring the angle with a protractor?' Wait 10 seconds; cold-call 3 learners. For Q3, invite a volunteer to show their Part C working on the board. If they used sin(110°) correctly, praise explicitly: 'Notice — sin of an obtuse angle is still positive. Well done for not switching to 70° without understanding why.' Write the decision rule in a box. Point back to the Kericho courtyard photograph: 'Look at the stage area. The contractor is standing outside — he cannot measure the height of the stage floor section.	Structured Peer Assessment with Metacognitive Prompting — the checklist gives learners explicit criteria language ('correct formula stated', 'correct substitution') that they internalise as self-monitoring habits. The whole-group Q&A then elevates individual procedural work to conceptual understanding of formula selection. Anchoring Q1 to the Kericho photograph re-connects the abstract exercise to the driving phenomenon.	Observable indicators: (i) peer checklist is filled with specific feedback, not just ticks; (ii) during Q1, learner can articulate 'the height is not measurable outdoors' rather than just 'we use the sine formula'; (iii) during Q3, learner correctly states sin(110°) ≈ 0.940 and confirms area ≈ 24.44 m². Red flag: learner who still cannot state the decision rule after the debrief — teacher pairs with a peer mentor for the model-building phase.

	measurable (e.g. folding fabric, tiling a floor already laid); USE SINE METHOD when only side lengths and an included angle are known (e.g. outdoor land, unmarked courtyard, farm plot).	The sine formula is his only practical tool.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board is displayed (physical board or shared screen). Learners write on sticky notes (or type into the shared DQB document) responses to two prompts: PROMPT 1 — 'What new tool do we now have for the courtyard problem that we did NOT have in Lesson 1?' PROMPT 2 — 'What part of the courtyard puzzle is STILL unanswered?' Learners post their notes in two columns: LEARNED and STILL WONDERING. The teacher reads 3–4 notes aloud and annotates the board: moves the parallelogram/rhombus formulas from the WONDERING column (posted in Lesson 3) to the LEARNED column, and highlights remaining items (e.g. 'How do we find the area of the hexagonal border tiles?', 'What about the triangular sections with no right angle?') to preview Lesson 6.	Hand out sticky notes (or open shared document). Say: 'You have 3 minutes — respond to both prompts. Be specific: name the formula and name the problem it solves in Kericho.' After posting, read 3 LEARNED notes aloud; for each, ask: 'Does the class agree this belongs in LEARNED?' Quick thumbs up/down. Then read 2 STILL WONDERING notes; say: 'Keep these in mind — our next lessons will address the hexagonal tiles and the triangular sections.' Point to the driving question written at the top of the board: 'How can we calculate the exact area of any polygon-shaped surface even when we cannot measure its height?' Ask: 'Are we closer to answering this? What percentage of the courtyard problem can we now solve?' Cold-call 2 learners for an estimate.	Public Knowledge Tracking (DQB Protocol) — moving items between WONDERING and LEARNED creates a visible, shared record of intellectual progress. The 'percentage solved' question gamifies progress and motivates engagement with remaining lessons. Highlighting STILL WONDERING items creates productive anticipatory curiosity about the pentagon and hexagon formulas.	Observable indicator: LEARNED sticky notes correctly name a specific formula (not just 'area') and link it to a specific courtyard problem (stage, kanga, garden). Red flag: learner writes a vague note ('we learned about shapes') — teacher prompts: 'Which formula? Which measurement problem in Kericho does it solve?'
Model Building Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for	Each group retrieves their cumulative Kericho courtyard scale-drawing model (A3 sheet from previous lessons). Using their Part A answer (≈ 36.25 m^2), learners write the area value neatly inside the parallelogram stage region on the drawing, add a colour-coded annotation box that states: 'Method used: Sine formula. Data: $a = 8$ m, $b = 5$ m, $\theta = 65°$. Area ≈ 36.25 m^2.' They also shade the stage region in a distinct colour (e.g. blue) to distinguish it from already-calculated triangular	Distribute models and say: 'You now have the stage area. Add it to your model — follow the annotation format on the board.' Write the format on the board: [Region name | Formula | Data | Area | Tiles needed]. Give groups 8 minutes. Circulate; check that the sine formula annotation is present (not just the number). After 8 minutes, cold-call one group to read their completion sentence aloud. Ask the class: 'Is this sentence mathematically complete? What could be	Cumulative Model Revision — physically annotating the scale drawing with calculated values, methods, and tile counts transforms an abstract exercise into a tangible engineering artefact. The tile-count extension connects the pure mathematical result to the contractor's real procurement decision, directly answering the driving question. The completion sentence is a writing-to-learn strategy that forces each group to articulate the conceptual breakthrough (sine	Observable indicators: (i) model annotation includes formula name, data, and area with correct units; (ii) tile count is correct (145–146) and the rounding-up decision is justified; (iii) completion sentence explicitly references 'sine formula' and 'included angle' rather than height. Red flag: area written without units (m^2) — correct immediately as a precision habit essential for the contractor scenario.

Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	sections (shaded in previous lessons). Groups then compute how many standard floor tiles (each 0.5 m × 0.5 m = 0.25 m²) would be needed to cover just the stage area: 36.25 ÷ 0.25 = 145 tiles, rounded up to 146 tiles to account for cuts at the edges. This number is written in a box at the edge of the model. Finally, each group writes one sentence on a label below their model: 'The contractor can now calculate the stage area without measuring the height because ___________.'	improved?' Wait 10 seconds. Collect one model per group for teacher review. Close by pointing to the courtyard photograph: 'Today you solved the stage. Next lesson we move to the hexagonal border tiles — a completely different polygon. Start thinking: what do you know about regular hexagons?'	formula removes the need to measure height).	

D. TEACHER REFLECTION
After the lesson, consider: (1) Did the Phase 1 tally show many learners defaulting to 'base × height' for Diagram A? If yes, the derive-phase lessons (3 & 4) may need re-teaching or a short re-engagement activity at the start of Lesson 6. (2) Were peer-assessment checklist comments substantive or superficial? If superficial, model a sample feedback comment at the start of Lesson 6 before any further peer work. (3) Did any group's tile count differ significantly from 146? Trace whether the error was in the area calculation or the division step — the two skills need separate remediation. (4) Were the STILL WONDERING DQB notes pointing toward hexagons and triangles (expected) or toward parallelogram content still unresolved (concern)? If the latter, insert a 5-minute recap at the start of Lesson 6 before introducing new polygon types. (5) Note which learners needed targeted support during Phase 3 (those who could not state the decision rule) — plan a brief check-in question for those learners at the start of Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined two diagrams of the Kericho courtyard parallelogram stage — one with an included angle and no height, one with a height and no angle — and predicted which formula applied to each. They also worked through a structured three-part problem set involving the courtyard stage (8 m × 5 m, 65°), a rhombus kanga patch from Nairobi's Gikomba market (diagonals 24 cm and 18 cm; also sides 15.23 cm and angle 56°), and a parallelogram school garden in Kisumu (6.5 m × 4 m, 110°), gathering numerical evidence that both formulas yield consistent results when both data sets are available.
What did I learn?	Learners learned that: (i) the sine area formula — Area = ab sin θ — is the practical choice when perpendicular height cannot be physically measured, as is the case for the Kericho courtyard stage; (ii) the diagonal formula — Area = ½ d_1d_2 — is efficient when both diagonals are accessible (e.g. on a flat fabric or already-laid tile); (iii) sin of an obtuse angle remains positive, so the formula works for angles greater than 90°; (iv) the two formulas are consistent — they yield the same area for the same shape when both data sets are correctly used; (v) the contractor needs to round tile counts up to avoid under-ordering material.
How does this explain the phenomenon?	Learners explained through peer assessment, whole-class discussion, DQB annotation, and model-building why the Kericho contractor must use the sine formula for the parallelogram stage: the courtyard surface has no accessible perpendicular height, but two side lengths and the included angle can be measured with a tape and protractor. They articulated a decision rule — use the diagonal method when diagonals are measurable; use the sine method when sides and an included angle are known — and applied it to judge which formula is more practical in each real-life Kenyan context. They updated the cumulative courtyard model

	with the stage area (≈ 36.25 m^2) and a tile-count estimate (146 tiles), advancing the class closer to fully answering the driving question.

LESSON 6 (80 minutes): Area of a Regular Pentagon: Deriving and Applying $A = (5s^2)/(4 \tan 36°)$

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, learners will be able to derive and apply the formula $A = (5s^2)/(4 \tan 36°)$ to calculate the area of a regular pentagon when only the side length is known, connecting this to the Kericho courtyard paving scenario.
Knowledge	Learners will know that: (i) the interior angle sum of a regular pentagon is 540°, making each interior angle 108°; (ii) a regular pentagon can be divided into 5 congruent isosceles triangles each with apex angle 72°; (iii) the apothem (height of each triangle) can be expressed as $(s/2)/\tan 36°$, leading to the formula $A = (5s^2)/(4 \tan 36°)$; (iv) tan 36° appears because the right-angled triangle formed by the apothem has a base of $s/2$ and a base angle of 36°.
Skills	Learners will be able to: (i) calculate interior angles of any regular polygon; (ii) decompose a regular pentagon into 5 congruent isosceles triangles and identify the apothem; (iii) derive $A = (5s^2)/(4 \tan 36°)$ step-by-step from first principles; (iv) apply the formula to find the area of a pentagonal njugu (groundnut) tray and other real-life pentagon-shaped objects; (v) evaluate whether the formula gives a result that is plausible in context.
Attitudes	Learners will appreciate that: (i) mathematical formulas are tools built from prior knowledge, not arbitrary rules to memorise; (ii) persisting through multi-step derivations builds confidence; (iii) sharing reasoning with peers strengthens collective understanding.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	The Kericho courtyard contractor has now measured the sides of the pentagonal njugu (groundnut) display tray that a market vendor uses outside the community hall — its shape will be replicated as a decorative inlay in the courtyard floor. The contractor knows only the side length of the regular pentagon tile but cannot physically measure its perpendicular height (apothem) because the tile is already grouted in a sample panel. In this lesson learners discover how to derive the exact area formula for a regular pentagon using only its side length, solving the contractor's problem and advancing the unit driving question.
Safety Notes	No hazardous materials are used. If physical pentagon tiles or card cutouts are handled, learners should take care with scissors or craft knives. Learners working in groups should pass shared materials carefully to avoid injury.

B. LESSON OVERVIEW

Lesson 6 is the Observe & Explain phase for the area of a regular pentagon. The lesson opens by presenting the market-vendor sub-phenomenon: a photograph of a pentagonal njugu (groundnut) display tray used by a vendor outside a community hall in Kericho, which the contractor wants to replicate as a decorative floor inlay. The teacher guides learners through a structured three-part investigation: (1) Predict — learners estimate and record their guesses for the area before any formula is introduced; (2) Observe — learners use a pre-drawn regular pentagon on 1 cm^2 grid paper to count squares and estimate area, then confirm interior angle calculations; (3) Explain — learners derive $A = (5s^2)/(4 \tan 36°)$ by dividing the pentagon into 5 congruent isosceles triangles, identifying the apothem using trigonometry, and working a guided example with a measured tile. The lesson closes with a DQB update centred on the question "Why does tan 36° appear in the pentagon formula?" Formative assessment is embedded throughout via mini-whiteboards, cold-calling, and a closing exit card.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Common and systematic naming: iso-, sec-, and tert- prefixes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a projected photograph of a pentagonal njugu (groundnut) tray used by a market vendor at Kericho town market. The tray is labelled with one measurement: each side is 18 cm long. Learners are told: 'The Kericho courtyard contractor wants to replicate this pentagon shape as a decorative tile inset in the floor. He needs to know the exact area of one tile so he can order the right amount of coloured cement paste.' Working individually and silently for 4 minutes, learners write on their mini-whiteboards: (a) their estimate of the area in cm²; (b) one method they think could work; (c) one reason why the method might fail. Pairs then share their predictions for 3 minutes. The class reconvenes and 4–5 predictions are shared aloud. Teacher records all predicted areas and methods on the DQB in a column headed 'Our Predictions — L6'.	Display the photograph clearly. Say: 'Look carefully at this njugu tray from Kericho market. Each side measures exactly 18 cm. The contractor needs its area. You have 4 minutes — write your estimate and your method on your whiteboard. Work alone and in silence.' Set a visible 4-minute timer. After 4 minutes say: 'Turn and share with your neighbour. You have 90 seconds.' After sharing, cold-call exactly 5 learners (choose a mix of confident and hesitant learners by name): 'Amina, what area did you estimate and why?' Apply 12-second wait time after each cold-call before accepting the answer or probing further. Record all five estimates verbatim on the board under 'Our Predictions.' Ask the class: 'Hands up — who thinks the base-times-height method will work here? Why might it fail?' Pause 10 seconds. Accept 2–3 responses. Say: 'Keep your whiteboards — we will come back to these predictions at the end of the lesson.'	Prediction Anchoring — by committing to an estimate before instruction, learners activate prior knowledge about area and expose their own misconceptions (e.g., trying to apply base × height directly). Recording predictions publicly on the DQB creates accountability and a visible benchmark against which the derived formula will be measured at the end of the lesson. This strategy is drawn from the NGSS Science and Engineering Practice of 'Asking Questions and Defining Problems.'	Observable indicator: every learner has written a numerical estimate and a named method on their whiteboard within 4 minutes. Teacher scans whiteboards during the pair-share and notes: (i) learners who write 'base × height ÷ 2' — these need scaffolding on why height is inaccessible; (ii) learners who attempt to measure the height from the photograph — affirm the instinct but surface the limitation; (iii) learners who leave the method blank — these are priority cold-call targets in Phase 2.
Observe Phase 📹 **VIDEO:** Common and systematic naming: iso-, sec-, and tert- prefixes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and	Each learner receives a Resource Sheet containing: (A) a pre-drawn regular pentagon with side length labelled s = 18 cm, printed on 1 cm² grid paper; (B) a blank table for recording observations; (C) a printed photograph of the Kericho courtyard scale drawing showing the pentagonal inlay region. TASK 1 (5 min, individual): Count grid squares inside the pentagon to estimate its area. Record in the table. TASK 2 (7 min, pairs): Using a protractor, measure all five interior angles of the printed	Distribute Resource Sheets. Say: 'Task 1 — count squares and estimate area. You have 5 minutes, individual, in silence.' After 5 minutes: 'Task 2 — measure every interior angle with your protractor. Work in pairs. 7 minutes.' Circulate; kneel beside any pair that is measuring incorrectly and say: 'Place the centre of your protractor exactly at the vertex — can you show me where that is?' After Task 2, cold-call 3 pairs for their angle sum: 'Ochieng and Fatuma — what did	Evidence-to-Pattern Reasoning (NGSS Practice 4: Analysing and Interpreting Data) — learners gather two independent pieces of evidence (grid-square count and direct angle measurement) before any formula is presented. The grid count gives an approximate numerical benchmark; the angle measurement reveals the 540° structure that will anchor the algebraic derivation. Connecting both to the Kericho courtyard photograph grounds the observation in the anchoring	Observable indicators: (i) each learner's resource sheet shows a grid-square count between 220 cm² and 240 cm² for s = 18 cm (the exact value is ≈ 222.5 cm²; estimates within ±15% are acceptable at this stage); (ii) each pair records an interior angle sum of 540° ± 5° measurement error; (iii) all learners correctly identify apex angle = 72° and base angle = 54° in the cold-call. If more than 3 learners cannot state the apex angle, pause and re-teach: 'We have 5 equal triangles sharing a

[Irregular Polygons (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	pentagon and record them. Calculate the angle sum. Divide by 5 to find each interior angle. Record in the table. TASK 3 (5 min, whole class discussion): Teacher projects a large regular pentagon and draws all 5 diagonals from the centre to each vertex, dividing it into 5 triangles. Learners observe the triangles and answer: 'What type of triangles are these? What is the apex angle of each triangle?' TASK 4 (3 min, individual): Learners record answers on their resource sheet and annotate the projected diagram.	you get for the sum?' Apply 12-second wait time. Affirm or correct: 'The sum must be 540°. If yours differs, what might have caused the error?' Project the pentagon divided into 5 triangles. Ask: 'What do you notice about the five triangles?' Wait 10 seconds. Cold-call 4 learners. Guide towards: 'They are congruent and isosceles.' Ask: 'What is the apex angle of each triangle? How do you know?' Wait 12 seconds. Cold-call 3 learners. Establish: 'Apex angle = 360° ÷ 5 = 72°.' Ask: 'So what are the two base angles of each isosceles triangle?' Wait 10 seconds. Establish: base angle = (180° − 72°) ÷ 2 = 54°.	phenomenon rather than treating it as abstract geometry.	central point — what is the full angle at the centre?'
Explain Phase 📹 **VIDEO:** [Common and systematic naming: iso-, sec-, and tert- prefixes](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	DERIVATION (12 min, teacher-led with learner participation): Learners follow the derivation on their resource sheet, filling in each blank step as the teacher works through it on the board. Step 1: State the area of one isosceles triangle in terms of base s and height h (apothem a): A_triangle = (1/2) × s × a. Step 2: Identify the right-angled triangle formed by drawing the apothem from the centre to the midpoint of one side. Its base = s/2, its angle at the base vertex = 36° (half of the apex angle 72°... wait — teacher pauses here and asks: 'Which angle should I use to find a in terms of s/2? The 36° or the 54°?'). Learners discuss in pairs for 90 seconds. Teacher cold-calls 4 learners. Establish: tan(54°) = a / (s/2) OR equivalently tan(36°) = (s/2) / a, so a = (s/2) / tan(36°). Step 3: Substitute a into A_triangle: A_triangle = (1/2) × s × (s/2)/tan(36°) = s²/(4 tan 36°). Step 4: Multiply by 5: A_pentagon = 5s²/(4 tan 36°). GUIDED	Say: 'We are going to build the formula step by step — nobody writes the formula until we derive it together.' Draw the right-angled half-triangle on the board. Label sides: hypotenuse (the equal leg of the isosceles triangle), base = s/2, height = apothem a, angle at base = 36°. Ask: 'In this right-angled triangle, which trig ratio connects a and s/2?' Wait 12 seconds. Cold-call 3 learners. After Step 2 derivation, pause and say: 'Turn to your neighbour — in your own words, explain why 36° appears here and not 54°.' Give 90 seconds. Cold-call 4 learners. Anticipated misconception: learners confuse the base angle 54° with the half-apex angle 36° — say: 'We drew the apothem to the MID-POINT of the side — which angle is at the midpoint vertex in our small right triangle?' For the guided example, do NOT reveal the calculator value of tan 36° immediately. Ask: 'Who can estimate tan 36° from our trig	Stepwise Co-Construction (NGSS Practice 6: Constructing Explanations) — the derivation is not presented as a finished product but built collaboratively, with learners filling in blanks and verbalising each logical link. The critical 'pause-and-discuss' moment at Step 2 (36° vs 54°) is a deliberate productive struggle point that forces learners to reason about which angle is in the correct right-angled triangle, directly explaining why tan 36° — and not tan 54° — appears in the formula. This directly answers the DQB update question for this lesson.	Observable indicators: (i) during co-derivation, all learners' resource sheets show correct entries at each blank step — teacher scans during the 90-second pair discussion; (ii) the guided example: all learners who call out steps use the correct substitution; (iii) independent application: target benchmark is ≈ 247.7 cm² for s = 12 cm — learners within ±2 cm² (accounting for rounding of tan 36°) meet expectations; learners with answers above 300 cm² or below 200 cm² receive immediate targeted questioning: 'Read me your substitution for s — what did you square?'

	EXAMPLE (7 min): Teacher works s = 18 cm step-by-step on the board, asking learners to call out each arithmetic step. $A = 5 \times 18^2 / (4 \times \tan 36°) = 5 \times 324 / (4 \times 0.7265) = 1620 / 2.9062 \approx 557.3$ cm². Compare with grid-count estimate. Discuss discrepancy and why the formula is exact. APPLICATION (6 min, individual): Learners solve: 'A second pentagon tile in the Kericho courtyard has side length 12 cm. Find its area to 1 decimal place.' Learners work independently, then peer-check.	tables?' Wait 10 seconds. After showing $A \approx 557.3$ cm², point to the DQB and say: 'How does this compare with your grid-count prediction? Keziah, what do you notice?' For the independent application (s = 12 cm), circulate and check: expected answer = $5 \times 144/(4 \times 0.7265) \approx 247.7$ cm². If a learner gets a very different answer, ask: 'Show me your substitution — where did each number come from in the formula?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Common and systematic naming: iso-, sec-, and tert- prefixes] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner receives two sticky notes. On the YELLOW note, they write: 'One thing I can now explain about pentagons that I could not at the start of Lesson 6.' On the BLUE note, they write: 'One question that this lesson raised for me.' Learners post their notes on the class DQB in the designated columns. The teacher reads 4–5 blue notes aloud and clusters them. The class discusses whether the question 'Why does tan 36° appear in the pentagon formula?' has been answered and records a class consensus response in one sentence on the DQB. Learners also update their personal 'Courtyard Progress Map' (a running diagram of the Kericho courtyard, with each area formula learned shaded in as it is derived) by shading the pentagonal inlay region and labelling it with the formula.	Distribute sticky notes. Say: 'You have 3 minutes — one yellow note, one blue note. Be specific: write a formula, a step, or a named concept — not just 'I learned area.'' After posting, read 4 blue-note questions aloud. Ask: 'Which of these questions did today's lesson answer?' Cold-call 3 learners. Then ask: 'Can we now answer the DQB question — Why does tan 36° appear in the pentagon formula? Who will give me a one-sentence answer?' Wait 12 seconds. Cold-call 2 learners. Aim for a class consensus sentence such as: 'Tan 36° appears because when we drop an apothem from the centre to the midpoint of a side, it creates a right-angled triangle whose base angle is half of 72° (the apex angle), which is 36°.' Write this sentence on the DQB. Say: 'Now shade the pentagon on your Courtyard Progress Map and write the formula.' Give 2 minutes.	DQB Collaborative Sensemaking (NGSS Practice 7: Engaging in Argument from Evidence) — the sticky-note protocol externalises individual understanding and publicly surfaces remaining uncertainty. Clustering blue-note questions reveals shared conceptual gaps, while the consensus sentence forces the class to articulate the mechanistic link between the geometry of the pentagon and the appearance of tan 36°. The Courtyard Progress Map makes the cumulative structure of the unit visible, reinforcing that each formula is a solution to a real measurement problem in the Kericho scenario.	Observable indicators: (i) yellow notes contain a specific mathematical statement (e.g., 'Each interior angle of a regular pentagon is 108°' or 'The apothem = (s/2)/tan 36°') — vague notes such as 'I learned area' are flagged and the learner is asked to add one specific detail before leaving; (ii) the class consensus DQB sentence correctly names 'half of 72°' or 'the right-angled triangle formed by the apothem' as the source of tan 36°; (iii) the Courtyard Progress Map shows the pentagon shaded and the correct formula written by all learners.
Model Building Phase	Learners open their cumulative 'Courtyard Model' — a hand-drawn or printed scale diagram of the	Say: 'Open your Courtyard Model. Today you add the pentagon tile areas. You have 7 minutes —	Cumulative Model Revision (NGSS Practice 2: Developing and Using Models) — each lesson	Observable indicators: (i) all learners record $A \approx 139.4$ cm² (±2 cm² for rounding) for s = 9 cm and

📹 **VIDEO:** Common and systematic naming: iso-, sec-, and tert- prefixes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Kericho courtyard that they have been annotating across all 8 lessons. Today's task: (a) Using the formula $A = 5s^2/(4 \tan 36°)$ and the courtyard scale drawing provided (where each pentagonal inlay tile has side = 9 cm at scale), calculate the area of one pentagonal inlay tile and record it on the model diagram. (b) The courtyard has 6 such pentagonal inlay tiles. Calculate the total area covered by pentagonal tiles. (c) On the model, write a speech bubble for the contractor: 'I can now find the area of the pentagon tile because...' — completing the sentence using mathematical language. Learners work independently for 7 minutes, then share with a shoulder partner for 3 minutes.	work independently.' Circulate. After 3 minutes, check: expected area for s = 9 cm: $A = 5 \times 81/(4 \times 0.7265) = 405/2.9062 \approx 139.4$ cm²; total for 6 tiles ≈ 836.2 cm². If a learner has a vastly different answer, ask: 'What value did you use for tan 36°? Show me the substitution.' After 7 minutes: 'Share your speech bubble sentence with your shoulder partner. Is the mathematical language accurate?' Cold-call 3 learners to read their speech bubble aloud. Probe: 'Wanjiku, you wrote the formula — can you explain in one sentence where the 4 in the denominator comes from?' Wait 12 seconds. Close the lesson: 'In Lesson 7 we move to the hexagonal border tiles on the courtyard. Think about this: what will be different when we divide a regular hexagon into triangles instead of a pentagon?'	adds a new layer of quantitative detail to the Kericho courtyard model, so the model functions as a living record of the unit's evidence-gathering sequence. The speech-bubble task requires learners to produce a causal explanation in their own words, which consolidates the derivation and connects the mathematics explicitly back to the contractor's real measurement problem. The closing bridge question ('What will be different for a hexagon?') primes learners for Lesson 7 and maintains the driving question as an open, advancing inquiry.	total ≈ 836.2 cm² on the model diagram; (ii) speech-bubble sentences include at least one of the following mathematical terms: 'apothem,' 'isosceles triangle,' 'tan 36°,' or '5 congruent triangles'; (iii) during cold-call of speech bubbles, learners can trace the origin of the '4' in the denominator to '$(1/2) \times s \times a$ where $a = (s/2)/\tan 36°$ gives $s^2/(4 \tan 36°)$' — if not, reteach the substitution step at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your professional journal:

1. DERIVATION COMPREHENSION: At the 'pause-and-discuss' moment (Step 2 — which angle, 36° or 54°?), how many learners could correctly justify the choice of 36° without prompting? If fewer than 60% could do so, plan a re-entry point at the start of Lesson 7 using a simpler right-triangle sketch.
2. COLD-CALL EQUITY: Review your cold-call record. Did you call on learners from all rows/ability groups? Did any learner receive more than 2 cold-calls? Were hesitant learners given the full 12-second wait time?
3. FORMULA VS. PROCEDURE: Did learners treat $A = 5s^2/(4 \tan 36°)$ as a formula to memorise, or as a result they derived? Evidence: check whether speech-bubble sentences reference the geometric origin (apothem, isosceles triangle) or merely restate the formula.
4. GRID ESTIMATE vs. FORMULA RESULT: For s = 18 cm, the grid estimate (Task 1) and the formula give ≈ 557.3 cm². Did learners make the connection between their estimate and the exact value? If the estimates were far off, consider providing more structured grid-counting guidance in future lessons.
5. DQB QUALITY: Were yellow notes specific enough? If more than 4 learners wrote vague statements, revise the sticky-note prompt for Lesson 7 to include a sentence starter: 'I can now calculate... because the formula shows that...'
6. COURTYARD PHENOMENON CONNECTION: Did learners consistently reference the Kericho courtyard and the njugu tray, or did the lesson drift into abstract formula work? If the phenomenon was lost, open Lesson 7 by revisiting the courtyard photograph before introducing hexagons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a regular pentagon (the njugu tray sub-phenomenon) printed on 1 cm^2 grid paper and counted squares to estimate its area. They measured all five interior angles with a protractor, confirmed the angle sum is 540° and each angle is 108°, and observed a projected diagram of the pentagon divided into 5 congruent isosceles triangles from the centre, identifying the apex angle as 72° and the base angles as 54°.
What did I learn?	Learners derived that the apothem (height) of each isosceles triangle equals (s/2) / tan 36°, because the apothem bisects the apex angle, forming a right-angled triangle whose base angle is 36°. Substituting into A = (1/2) × base × height for one triangle and multiplying by 5 yields $A = 5s^2/(4 \tan 36°)$. For s = 18 cm, A ≈ 557.3 cm^2; for s = 9 cm (courtyard inlay scale), A ≈ 139.4 cm^2. Tan 36° appears specifically because the right-angled half-triangle has a base angle of 36° — half of the 72° apex angle.
How does this explain the phenomenon?	Learners explained, in speech-bubble sentences on their Courtyard Model, that the contractor can find the exact area of a regular pentagon tile using only its side length s, because dropping a perpendicular from the centre to the midpoint of any side creates a right-angled triangle in which the base is s/2 and the base angle is 36°, making the apothem equal to (s/2)/tan 36°. The full pentagon area then follows by calculating the area of one of the 5 congruent triangles and multiplying by 5. The DQB was updated with the class consensus answer to 'Why does tan 36° appear in the pentagon formula?'

LESSON 7 (80 minutes): Area of a Regular Hexagon — Deriving the Formula from Equilateral Triangles	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive the formula $A = (3\sqrt{3}/2)s^2$ for a regular hexagon by decomposing it into six equilateral triangles, then apply it to calculate the total area of hexagonal paving tiles and honeycomb cells — completing the formula toolkit needed to solve the Kericho courtyard problem.
Knowledge	A regular hexagon can be divided into exactly 6 congruent equilateral triangles; the area of each equilateral triangle is $(\sqrt{3}/4)s^2$, giving a total hexagon area of $A = (3\sqrt{3}/2)s^2$; this formula requires only the side length s, with no perpendicular height needed.
Skills	Decompose a regular hexagon into equilateral triangles; derive $A = (3\sqrt{3}/2)s^2$ algebraically; calculate the area of a single hexagonal tile and a field of tiles; connect the hexagon formula back to the triangle-area formula from Lesson 1; update the Driving Question Board with final answers.
Attitudes	Appreciate that mathematical structures are deeply connected — the sine-area triangle formula from Lesson 1 underlies every formula developed across this unit; demonstrate persistence in algebraic derivation; share reasoning responsibly within group work.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	The Kericho courtyard contractor now has all three missing tools: the sine-area triangle formula (Lesson 1), Heron's formula (Lesson 2), the parallelogram formula (Lesson 3), the pentagon formula (Lesson 6), and now the hexagon formula. With $A = (3\sqrt{3}/2)s^2$ in hand, the contractor can calculate exactly how many hexagonal border tiles are needed without ever measuring a height. This lesson also closes the storyline arc by explicitly tracing how every formula in the unit descends from the equilateral-triangle area — the same triangle formula introduced in Lesson 1.
Safety Notes	No chemicals or sharp tools used. If physical honeycomb frames are brought in, warn learners not to consume wax or any residue without confirmation it is food-safe. Ensure learners with bee/honey allergies are seated away from any physical honeycomb sample.

B. LESSON OVERVIEW
Lesson 7 is the Observe & Explain phase for the regular hexagon. The lesson opens by re-anchoring learners to the Kericho courtyard phenomenon — specifically the hexagonal border tiles whose side length the contractor knows but whose height cannot be measured. Learners first predict the area of a hexagonal tile by inspection, then physically observe and decompose hexagonal objects (pencil cross-sections traced on paper, printed hexagonal grid sheets, and a diagram of a honeycomb frame) into six equilateral triangles. Using the equilateral-triangle area formula $A = (\sqrt{3}/4)s^2$ — which they connect explicitly to the sine-area formula from Lesson 1 — learners derive $A = (3\sqrt{3}/2)s^2$ for the full hexagon. They then apply the formula to (a) calculate the area of one courtyard border tile and (b) calculate the total wax area of a honeycomb frame with a given number of cells and cell side length. The lesson closes with a full DQB update: every opening question from the unit is revisited and answered, and learners articulate how the unit's formulas form a connected family rooted in the triangle. Assessment is through whiteboard derivation checks, a cold-call relay, and a short written application problem.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Equivalent fraction models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a photograph of the Kericho courtyard scale drawing (re-displayed from Lesson 1) with the hexagonal border tile highlighted in yellow. Each learner receives a printed regular hexagon outline (side length labelled s = 8 cm) and is asked to write a prediction: 'Without measuring the height of this hexagon, estimate its area in cm^2. Explain your reasoning in one sentence.' Learners write individually for 3 minutes, then share predictions with a shoulder partner. Pairs record the range of their two guesses on a sticky note posted on the class 'Prediction Wall'. Predictions are NOT corrected yet — they are left visible to be revisited after derivation.	Display the courtyard photograph and say: 'The contractor in Kericho knows every hexagonal border tile has a side length of 8 cm. The tiles are packed tightly — no height is visible or measurable. Your job right now is to make your best mathematical guess at the area of ONE tile — and write down your reasoning.' Distribute the hexagon outline sheet. Allow 3 minutes of silent individual thinking. After 3 minutes, say: 'Share your prediction with your shoulder partner. Agree on a range — what is the smallest reasonable answer and the largest?' WAIT 2 minutes for pair discussion. Cold-call 4 pairs (select one high-confidence pair, one low-confidence pair, one pair that drew on the sheet, one pair that used a formula attempt). Ask each: 'What did you estimate, and what mathematical reasoning did you use?' Do NOT evaluate answers — respond with 'Interesting — hold that thought.' Post sticky notes on the Prediction Wall.	Prediction Wall — making prior knowledge visible. By publicly posting predictions before instruction, learners activate relevant prior knowledge (area formulas they already know) and create a personal stake in discovering the correct formula. The visible spread of predictions also surfaces misconceptions (e.g., 'I used base × height but guessed a height') that the Observe phase directly addresses.	Observable indicator: Can the learner articulate a mathematical strategy (even an imperfect one) for estimating hexagon area? Listen for: reference to triangles, use of a guessed height, or attempts to apply rectangle/parallelogram ideas. Red flag: learner leaves the sheet blank or writes 'I don't know' — pair that learner with a strong partner for the Observe phase and provide the prompt: 'How many triangles do you think you could fit inside this shape?'
Observe Phase 📹 **VIDEO:** [Equivalent fraction models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Regular and Irregular Polygons (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Decomposition with physical objects (20 min): Each group of 4 receives: one sharpened hexagonal pencil, one sheet of plain paper, a ruler, and a protractor. Learners roll the pencil on the paper and trace its hexagonal cross-section, producing a regular hexagon. They then use a ruler to draw all three long diagonals (connecting opposite vertices), dividing the hexagon into 6 triangles. Learners measure the sides of each triangle and all internal angles to confirm: every triangle is equilateral and all six are congruent. They record measurements in a table: Triangle	Distribute materials and say: 'Step 1 — trace the pencil cross-section carefully. Press firmly and rotate slowly to get a clean hexagon.' Allow 4 minutes. Then say: 'Now draw a line from each vertex to the opposite vertex — you should get exactly 3 lines. What shapes have you created inside?' WAIT 10 seconds. Cold-call 3 learners: 'Amina — how many triangles? Korir — are they the same size? Atieno — what kind of triangles are they?' After confirmation, say: 'Measure every side and every angle of each triangle and complete the table. Take 8 minutes.' Circulate and prompt:	Decomposition-and-Measurement Strategy — learners build understanding by physically breaking a complex shape into familiar components (equilateral triangles) they already know how to handle. The honeycomb context makes the observation concrete and Kenyan-relevant. The deliberate link to Lesson 1 reinforces the unit's central thread: all polygon area formulas in this unit are extensions of the sine-area triangle formula, preventing fragmented memorisation of isolated rules.	Observable indicators: (1) The completed measurement table shows all six triangles with sides ≈ equal and all angles ≈ 60° — acceptable tolerance ± 2 mm, ± 2°. (2) The learner correctly writes $A\_equilateral = (\sqrt{3}/4)s^2$ from the sine-area formula. Red flags: (a) Learner draws diagonals incorrectly (e.g., draws only 2 diagonals) — intervene with 'How many vertices does a hexagon have? Connect each vertex to the one directly opposite.' (b) Learner cannot connect to Lesson 1 — provide the printed Lesson 1 formula strip as a scaffold.

	| Side 1 | Side 2 | Side 3 | Angle A | Angle B | Angle C. Part B — Honeycomb diagram (10 min): Learners examine a printed diagram of a honeycomb frame (labelled 'Typical bee colony frame, Kericho highlands apiary') showing a 5 × 8 grid of regular hexagonal cells, each with side length $s = 1.2$ cm. They annotate the diagram by outlining one cell, then subdividing it into 6 equilateral triangles — mirroring what they did with the pencil tracing. They count total cells (40) and note the side length. Part C — Connecting to Lesson 1 (5 min): Learners open their exercise books to Lesson 1 notes and locate the sine-area formula: $A_{triangle} = \frac{1}{2}ab \sin C$. They write the specialisation for an equilateral triangle ($a = b = s$, $C = 60°$): $A_{equilateral} = \frac{1}{2} \cdot s \cdot s \cdot \sin 60° = (\sqrt{3}/4)s^2$. They verify this numerically for $s = 8$ cm on a calculator.	'How do you know all six triangles are congruent? What property of the regular hexagon guarantees this?' After the table is complete, direct attention to the honeycomb diagram: 'This is a real honeycomb frame from a Kericho highlands apiary — the same county as our courtyard. Subdivide one cell into triangles exactly as you just did with the pencil.' Allow 5 minutes. Then say: 'Now go back to Lesson 1 in your notes. You derived a formula for the area of any triangle given two sides and an included angle. Write that formula, then substitute $a = b = s$ and $C = 60°$. What do you get?' WAIT 15 seconds. Cold-call 2 learners to write their result on the board.		
Explain Phase 📹 **VIDEO:** Equivalent fraction models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Algebraic derivation (15 min): Working in pairs, learners derive the hexagon area formula step by step in their exercise books, guided by a structured scaffold on the board: Step 1 — Write the area of ONE equilateral triangle with side s. Step 2 — Write the area of SIX such triangles. Step 3 — Simplify to $A = (3\sqrt{3}/2)s^2$. Step 4 — Verify numerically: substitute $s = 8$ cm and calculate A on a calculator; compare with their Lesson 1 prediction. Each pair checks their derivation against the pair next to them before the teacher cold-calls. Part B — Application to the Kericho courtyard (10 min): Learners read the problem from a	Write the three-step scaffold on the board and say: 'Use the result you just derived in the Observe phase to complete each step. Do NOT look up the formula — build it from what you already know.' WAIT 12 minutes for pair work. Circulate, asking: 'Show me Step 2 — what is $6 \times (\sqrt{3}/4)s^2$? Can you simplify that fraction?' After 12 minutes, run a cold-call relay: call Learner 1 to read Step 1, Learner 2 to read Step 2, Learner 3 to simplify Step 3, Learner 4 to give the numerical answer for $s = 8$ cm. Correct errors in real time. Then distribute the courtyard application card and say: 'This is the actual problem the Kericho contractor faces. Solve all three parts. You	Structured Algebraic Relay — the three-step scaffold prevents learners from copying a formula without understanding its origin, while the relay cold-call makes each derivation step a shared public act. The courtyard application immediately grounds the abstract formula in the phenomenon, and the adhesive-box rounding question introduces a real-world constraint (always round up when ordering materials) that connects mathematics to practical decision-making. The honeycomb extension completes the sub-phenomenon arc introduced in the Observe phase.	Observable indicators: (1) Correct derivation: $A = 6 \times (\sqrt{3}/4)s^2 = (6\sqrt{3}/4)s^2 = (3\sqrt{3}/2)s^2$ — all three algebraic steps present. (2) Courtyard answers: (a) $A_{one\ tile} = (3\sqrt{3}/2)(12^2) \approx 374.1$ cm²; (b) Total ≈ 17 957 cm²; (c) 17 957 ÷ 3 000 ≈ 5.99 → 6 boxes. (3) Honeycomb: $A_{total} = 40 \times (3\sqrt{3}/2)(1.2^2) \approx 149.6$ cm². Red flags: learner uses base × height with a guessed height — redirect: 'Where did you get that height? Can you measure it on a regular hexagon without extra information?' Learner forgets to round up boxes — ask: 'If you order 5 boxes, will the job be finished?'

	printed card: 'The hexagonal border of the Kericho community hall courtyard requires 48 tiles. Each tile has a side length of 12 cm. (a) Calculate the area of one tile. (b) Calculate the total area covered by all border tiles. (c) If each box of tile adhesive covers 3 000 cm², how many boxes must the contractor order?' Learners solve individually, then compare answers in their group of 4. Part C — Honeycomb wax area (5 min): 'The honeycomb frame has 40 cells, each with side length 1.2 cm. Calculate the total wax-boundary area of all cells.' (Note: this extends the observation phase calculation and rewards learners who correctly identified 40 cells.)	have 10 minutes.' After 10 minutes, ask: 'Who got a different answer for part (c) — the number of boxes? Tell me your answer.' WAIT 10 seconds. Cold-call 3 learners. Discuss rounding up: 'In real construction, do we round up or down? Why?' For the honeycomb extension, say: 'Use the same formula. Work fast — you have 5 minutes.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Equivalent fraction models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives a set of colour-coded sticky notes (green = answered, yellow = partially answered, orange = new question arising). The DQB from Lesson 1 is displayed at the front, showing all opening questions from the unit. Groups review every question on the board and decide: Which questions can now be fully answered? Which are partially answered? Do any new questions arise from today's lesson? Groups write final answers on green sticky notes and replace the original question cards. The class then nominates one spokesperson to read the full DQB aloud as a unit summary, constructing a verbal narrative of the complete storyline: 'We started with a courtyard in Kericho where no height could be measured. In Lesson 1 we learned... In Lesson 7 we finally...' Each learner writes a personal DQB exit sentence: 'The formula $A = (3\sqrt{3}/2)s^2$ helps the Kericho	Display the original DQB and say: 'This board has been with us since Lesson 1. Look at every question. For each one, decide — green: fully answered; yellow: partly answered; orange: new question.' Allow 6 minutes for group discussion. Circulate and listen for: are learners making explicit connections across lessons? After 6 minutes, call the class to attention: 'I need one spokesperson from each group to place your sticky notes. Briefly say which question you answered and how.' Facilitate placement — 4 minutes. Then say: 'I want one volunteer to narrate the whole storyline from Lesson 1 to Lesson 7 in 90 seconds — like telling a story to the contractor in Kericho.' WAIT 10 seconds. Select a mid-ability learner (not always the highest performer). After the narration, say: 'Now write your personal exit sentence: $A = (3\sqrt{3}/2)s^2$ helps the Kericho	Storyline Narration and DQB Closure — having a learner narrate the unit arc as a coherent story (not a list of formulas) activates narrative cognition, which research shows improves retention of conceptual connections. The colour-coded sticky-note protocol makes the cumulative progress of understanding visible and tangible. The personal exit sentence requires each learner to articulate the phenomenon-to-formula connection in their own words, revealing depth of understanding.	Observable indicators: (1) Green sticky notes contain accurate, formula-based answers (not vague statements). (2) The 90-second narration includes: phenomenon → problem (no height) → triangle formula (Lesson 1) → Heron's (Lesson 2) → parallelogram (Lesson 3) → pentagon (Lesson 6) → hexagon (Lesson 7). (3) Exit sentence explicitly states WHY the formula is useful for the contractor (e.g., 'because only the side length s of each tile is known, and the formula uses only s'). Red flag: exit sentence says only 'because it gives the area' — prompt: 'What information does the contractor have? What information does the formula need?'

	contractor because...'	contractor because...' WAIT 3 minutes for silent writing. Collect exit sentences as a formative assessment artifact.		
Model Building Phase 📹 **VIDEO:** Equivalent fraction models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Regular and Irregular Polygons (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative courtyard model (the annotated scale drawing they have been building since Lesson 1). They add the final layer: (a) Label the hexagonal border region with the formula $A = (3\sqrt{3}/2)s^2$ and write in the calculated total border area. (b) Add a 'Formula Family Tree' in the margin — a diagram showing how all five formulas derived in the unit descend from A_triangle = ½ab sin C: triangle (L1) → Heron's (L2) → parallelogram (L3) → regular pentagon (L6) → regular hexagon (L7). (c) Write a final contractor's note at the bottom of the model: 'Total materials needed: [triangle sections area] + [parallelogram stage area] + [hexagonal border area] = [Grand Total]. Recommended orders: tiles = ..., adhesive boxes = ..., budget = ...' Learners who finish early extend by calculating what percentage of the total courtyard area is covered by hexagonal tiles.	Distribute the cumulative model drawings and say: 'This is the last lesson in our evidence sequence. Add the hexagon layer to your model — formula, calculated area, and the Formula Family Tree. You have 10 minutes.' Circulate and observe: Is the Formula Family Tree connecting formulas correctly? Is the Grand Total calculation consistent with values from previous lessons? After 8 minutes, say: 'Now write the contractor's final note at the bottom. Imagine you ARE the contractor presenting this to the Kericho County council — make it clear, accurate, and professional.' WAIT for final 2 minutes. Close the lesson by holding up a completed model and saying: 'In Lesson 1 we could not calculate anything because we had no height. Today we completed the last formula the contractor needed. The courtyard can now be paved correctly — and no material will be wasted.' Ask the class: 'How many different area formulas did we develop across this unit?' WAIT 10 seconds. Cold-call 3 learners for the count and the names.	Cumulative Model Revision with Formula Family Tree — the Family Tree is a knowledge-structure diagram (a form of concept mapping) that makes the mathematical relatedness of all five formulas explicit and visual. Research on schema formation shows that learners who explicitly map relationships between formulas retain and transfer them more reliably than those who treat each formula as isolated. The contractor's final note bridges abstract mathematics and professional communication, addressing the KICD competency of applying mathematics to real-life decisions.	Observable indicators: (1) Formula Family Tree correctly shows all five formulas as branches from A = ½ab sin C. (2) Grand Total is present and arithmetically consistent with previous lessons' sub-totals. (3) Contractor's note states a specific tile count, adhesive box count, and a cost estimate (even if approximate). Red flag: Formula Family Tree is missing or shows formulas as unconnected — sit with the learner and ask: 'The hexagon formula — where did we start to derive it today? And where did that triangle formula come from in Lesson 1?' Trace the chain verbally before asking them to draw it.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Could all learners complete the algebraic derivation of $A = (3\sqrt{3}/2)s^2$ independently, or did most rely on the scaffold? If the majority needed the scaffold throughout, plan a consolidation starter for Lesson 8 in which learners re-derive the formula from memory in 3 minutes. (2) Were the DQB green sticky notes substantive (formula-based answers) or vague? If vague, the exit sentences will reveal whether the gap is conceptual or communicative — read them tonight and identify 3–5 learners who need a targeted follow-up question at the start of Lesson 8. (3) Did the 90-second storyline narration include all five formula milestones, or did learners skip the pentagon (Lesson 6)? If the pentagon was omitted, the unit arc is not yet coherent for those learners — address in the opening of Lesson 8. (4) Was the honeycomb extension completed by more than half the class? If yes, consider adding a further extension in Lesson 8 comparing honeycomb efficiency (hexagon vs. square vs. triangle packing). (5) Note which learners could not connect $A = (3\sqrt{3}/2)s^2$ to $A = ½ab \sin C$ — these learners may struggle with the Lesson 8 synthesis task; prepare a bridging worked example pairing both formulas side by side.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners traced hexagonal pencil cross-sections and honeycomb cell diagrams, subdivided them into 6 equilateral triangles, measured sides and angles to confirm congruence, and recorded data in a structured table — providing direct physical evidence that every regular hexagon is composed of 6 identical equilateral triangles.
What did I learn?	A regular hexagon of side length s has area $A = (3\sqrt{3}/2)s^2$ — derived by multiplying the equilateral triangle formula $A = (\sqrt{3}/4)s^2$ by 6. This formula requires only the side length and no perpendicular height, making it directly applicable to the Kericho courtyard hexagonal border tiles where no height can be physically measured.
How does this explain the phenomenon?	Learners explained why the hexagon formula is a direct descendant of the sine-area triangle formula from Lesson 1 ($A = ½ab \sin C$, specialised to equilateral triangles), constructed a Formula Family Tree linking all five polygon area formulas developed across the unit, applied the hexagon formula to calculate tile area and adhesive box quantities for the Kericho courtyard, and updated the DQB with final answers — completing the full evidence-gathering arc of the unit.

LESSON 8 (80 minutes): Synthesis, Model Building & Final Assessment: The Complete Kericho Courtyard	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all area-of-polygons formulas developed across the unit, apply them to the complete Kericho courtyard scenario, and demonstrate mastery through group presentation, whole-class model revision, and an individual written portfolio task.
Knowledge	Learners will integrate knowledge of: (a) area of a triangle using two sides and an included angle (sine formula); (b) area of a triangle using three sides (Heron's Formula); (c) area of a rhombus and parallelogram; (d) area of a regular pentagon and hexagon — applying each formula correctly to the appropriate section of the Kericho courtyard floor plan.
Skills	Learners will: compute total courtyard area by summing all polygon sub-areas; select and justify the appropriate formula for each section; annotate a scale drawing with computed areas; present calculations with mathematical reasoning; and individually complete a written class activity that assesses all five SLOs for Sub-Strand 2.4.
Attitudes	Learners will appreciate how mathematical formulas are practical tools — not abstract exercises — that enable builders, farmers, and designers in communities like Kericho to make accurate material-ordering decisions, avoiding waste and financial loss.
Key Inquiry Question	How are areas of polygons useful?
Purpose in Storyline	This is the capstone lesson. Learners return to the full Kericho community-hall courtyard — the anchoring phenomenon introduced in Lesson 1 — now armed with every formula developed across the unit. They calculate every section's area (triangular sections via sine formula and Heron's, the parallelogram stage, the hexagonal border tiles), sum them to a total, and determine how many tiles and how much paving material the contractor must order. Groups present and justify their choices, the class revises the cumulative courtyard model together, and learners individually demonstrate mastery through a portfolio assessment task. The Driving Question Board is formally closed as learners articulate how each lesson answered one of their original questions.
Safety Notes	No physical hazards. Remind learners to handle rulers, geometrical sets, and calculators with care. Ensure all learners have adequate workspace during the individual written assessment to uphold academic integrity.

B. LESSON OVERVIEW

In this final lesson of Sub-Strand 2.4, learners synthesise all five polygon area formulas developed across Lessons 1–7 by applying them to every section of the Kericho community-hall courtyard — the anchoring phenomenon for the entire unit. The lesson opens with a brief Predict phase in which learners estimate the total courtyard area and predict which formula(s) they will need most. Groups then work through the complete courtyard scale drawing (Observe/Apply phase), computing areas of all triangular sections (sine formula and Heron's), the parallelogram stage, and the hexagonal border tiles, before summing to a total material requirement. Each group presents and justifies their formula choices to the class (Explain phase). The teacher then facilitates a whole-class model revision on the projected courtyard drawing, comparing the Lesson-1 naive model (base × height only) with the final annotated model (all five formulas). The Driving Question Board is reviewed and formally closed. The lesson concludes with an individual written class-activity portfolio task that assesses all five SLOs for Sub-Strand 2.4, providing summative evidence of mastery.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a clean copy of the full Kericho courtyard scale drawing (all sections labelled: Triangle A, Triangle B, Triangle C, Parallelogram Stage, Hexagonal Border). Working individually for 4 minutes, they write in their exercise books: (1) their estimate of the total courtyard area in m²; (2) which formula they will use for each labelled section and why; (3) one thing they were confused about in Lesson 1 that they now understand. Learners then share predictions with a shoulder partner for 2 minutes before 4 individuals share with the class.	Distribute the full courtyard scale drawing. Say: 'Before we calculate anything, I want to know what your mathematician's instinct tells you. Look at every section of this courtyard. In your book, write your estimate of the total area, the formula you will use for each section, and one thing about area of polygons that confused you in Lesson 1 but is now clear. You have 4 minutes — work silently and independently.' [START TIMER. WAIT 4 minutes.] After writing: 'Turn to your shoulder partner. Compare your formula choices. Where do you disagree? You have 2 minutes.' [WAIT 2 minutes.] Cold-call 4 learners — select at least one who estimated high, one who estimated low. Ask: 'Amina, which formula did you choose for the hexagonal border and why?' [WAIT TIME 12 seconds after each response.] Record predictions on the board in a two-column table: SECTION | PREDICTED FORMULA. Do NOT confirm or correct yet. Say: 'Let us keep these predictions visible. By the end of this lesson, you will know exactly how accurate your instincts are — and more importantly, why.'	Activation of Prior Knowledge / Prediction Accountability. By committing predictions to writing before calculating, learners engage metacognitive reflection on their own learning journey across the unit. The shoulder-partner comparison surfaces disagreements that motivate careful formula justification during the group work phase. The visible prediction table on the board creates cognitive accountability — learners are motivated to revisit and correct their own predictions, deepening retention.	Observable indicator: Can each learner correctly name the appropriate formula (sine rule area, Heron's, parallelogram, hexagon) for each courtyard section without prompting? Listen during cold-calls for correct formula names and reasoning (e.g., 'I use the sine formula for Triangle A because I have two sides and the included angle but no perpendicular height'). Flag any learner who defaults to 'base × height ÷ 2' for a triangle with no right angle — this signals a misconception to address during group work.
Observe Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Learners work in pre-assigned groups of 4–5. Each group receives the full courtyard data sheet containing all measurements needed: Triangle A (two sides and included angle), Triangle B (three sides only), Triangle C (two sides and included angle), Parallelogram Stage (base and included angle, or base and height), and the Hexagonal Border (side length of each tile and number of tiles). Groups divide sub-tasks: each	Distribute data sheets and A3 paper. Say: 'The contractor in Kericho is waiting. Your group must calculate the area of every single section of this courtyard using the correct formula — the one that actually works when you cannot drop a perpendicular. Divide the sections among yourselves. Every calculation must show: formula stated, values substituted, final answer in m² with two decimal places. You have 20	Jigsaw Accountability within Groups / Contextualised Problem Solving. Dividing sections among group members ensures every learner actively applies at least one formula rather than observing. The real-world cost and tile-count extensions embed proportional reasoning and connect area to genuine contractor decisions, reinforcing the unit's core message: these formulas exist to solve real material-ordering	Observable indicators during circulation: (1) Learners correctly state the full formula before substituting values — not just writing numbers. (2) For Heron's formula, learners compute $s = (a+b+c)/2$ correctly before computing the square root expression. (3) For the sine-area formula, learners correctly identify the included angle (not just any angle). (4) For the hexagon, learners use Area = $(3\sqrt{3}/2)s^2$ or

Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	member is responsible for at least one section's calculation. Groups must show all working (formula stated → values substituted → answer with units), complete a group summary table (Section | Formula Used | Calculated Area), and compute the total area. Groups also answer: 'The contractor charges Ksh 3,200 per m² for tiling. What is the total cost?' and 'If each hexagonal tile covers exactly the area you calculated, how many tiles are needed for the border?' Groups record everything on A3 paper for the presentation.	minutes.' [START TIMER.] Circulate continuously. At the 5-minute mark, check that every group has correctly identified the formula for each section before calculations proceed. Targeted prompts — do NOT give answers: For Triangle B (Heron's): 'What is your first step before you can use Heron's formula?' [WAIT 10 seconds.] 'What is s?' For Triangle A: 'You have written ½ab sin C. What are a, b, and C in this triangle specifically?' For the hexagon: 'What is the formula for a regular hexagon with side length s? How did we derive it in Lesson 7?' For the parallelogram: 'Can you use both the sine method and the base-times-height method here? Which is more direct given the data you have?' At 15 minutes, give a 5-minute warning. At 20 minutes, say: 'Finalise your summary table and total area. Write your total in large figures at the top of your A3 sheet — the contractor needs to see it clearly.' If a group finishes early: 'Calculate the percentage of total area covered by: (a) the triangular sections combined; (b) the hexagonal border. What does that tell you about which section needs the most tiles?'	problems in communities like Kericho. The A3 summary table makes thinking visible and ready for peer scrutiny during presentations.	the equivalent derived form from Lesson 7. (5) All areas are in m² with appropriate decimal precision. Flag groups whose total area falls outside a reasonable range (announce expected range: 'Your total should be between 80 m² and 140 m² — if you are outside this range, recheck your unit conversions from the scale drawing.').
Explain Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Two or three groups (selected by the teacher based on contrasting approaches observed during Phase 2) present their A3 summaries to the class. Each presenting group has 3 minutes: one member states the formula used for each section and why it was the right choice; another member reads out the total area and the contractor cost. Listening groups use a peer-feedback card	Select 2–3 groups that used contrasting approaches (e.g., one group used the sine method for the parallelogram, another used base × height — both correct). Say: 'Group 2, you are first. One person explains formula choices — section by section. Another person reads the total and the cost. The rest of the class listens with your peer-feedback card.' [TIMER: 3 minutes per group.] After each	Structured Peer Critique / Evidence-Based Argumentation. Requiring presenting groups to justify formula choice — not just state answers — elevates discourse from procedural reporting to mathematical reasoning. The peer-feedback sentence starters scaffold constructive critique and mirror scientific practice of evaluating evidence-based claims. Returning	Observable indicators: (1) Presenting learners can articulate WHY a formula is appropriate (e.g., 'We used Heron's formula for Triangle B because we were given all three sides but no angle and no height'). (2) Listening learners' peer-feedback cards contain specific mathematical references, not vague praise. (3) When discrepancies arise between groups' totals, learners can identify

[Area Models for Decimal Multiplication (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	with two sentence starters: 'We agree with your formula choice for ___ because ___' and 'We would question your calculation for ___ because ___.' After presentations, the full class compares totals and resolves any discrepancies. The teacher then returns to the prediction table from Phase 1 and invites learners to tick correct predictions and cross incorrect ones, narrating what changed in their thinking.	presentation, facilitate peer feedback: cold-call one listener group — 'Group 4, do you agree with their formula for Triangle B? Why or why not?' [WAIT 12 seconds.] If a group presents an error, do NOT immediately correct — ask the class: 'Does everyone agree with that result? What would you check first?' [WAIT 10 seconds, then cold-call.] After presentations, return to the prediction table: 'Look at your prediction from Phase 1. Put a tick next to every formula choice you got right. Put a question mark next to anything that surprised you. Turn to your neighbour and say one thing that changed in your thinking since Lesson 1.' [WAIT 2 minutes.] Cold-call 3 learners on what changed. Anchor each response back to the phenomenon: 'And that is exactly the insight that allows the Kericho contractor to order the right amount of material without ever being able to measure a perpendicular height.'	to the Phase 1 prediction table makes learning visible as a trajectory, helping learners see how their understanding has grown across the unit rather than treating each lesson in isolation.	the source of error (unit conversion, wrong formula, arithmetic). (4) At least 80% of learners can correctly tick their formula predictions — if fewer can, pause and re-teach formula selection criteria before moving to the model-building phase.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons](#) Source: CK-12 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)](#)	The teacher projects the Driving Question Board that has been updated across all 8 lessons. Learners take 3 minutes to review all posted questions and answer cards silently. Each learner then writes in their book: (a) the one question from the DQB that they personally found hardest to answer and why; (b) a one-sentence answer to the unit's driving question in their own words. A full-class share follows. The DQB is formally closed: learners read aloud the driving question one final time, then the teacher reveals the annotated courtyard drawing with all correct	Project the full DQB. Say: 'This board has been with us since Lesson 1. Every sticky note represents a question you or your classmates asked. Take 3 minutes — silently read every question and its answer card. Find the one question that was hardest for you personally to understand.' [WAIT 3 minutes.] Cold-call 5 learners: 'Kipchoge, which question was hardest for you and why?' [WAIT 12 seconds.] Then: 'Now, in one sentence in your book, answer the unit driving question: How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its	DQB Closure / Narrative Arc Completion. Formally closing the DQB provides a powerful metacognitive moment — learners experience the satisfaction of a complete explanatory arc, from puzzlement in Lesson 1 ('how can we find area without a height?') to confident multi-formula mastery in Lesson 8. Asking learners to write a one-sentence answer to the driving question in their own words consolidates understanding at the highest level of Bloom's taxonomy (synthesis/evaluation) and produces a memorable artefact of their learning. Revealing the annotated courtyard drawing	Observable indicators: (1) Learners' one-sentence driving-question answers explicitly reference at least two formulas and the concept of 'without needing a perpendicular height.' Collect 5–6 books to scan during the assessment phase. (2) During cold-calls, learners can articulate a complete, coherent answer — not just name a formula. (3) Any learner who cannot identify which DQB question was hardest (indicating they have not engaged with the board across the unit) should be noted for additional support during or after the portfolio task.

Source: CK-12 🔍 Search ARES for similar readings	areas filled in — the completed answer to the phenomenon that opened Lesson 1.	height — so that builders, farmers, and designers never order too much or too little material?' [WAIT 2 minutes for writing.] Cold-call 4 learners to share their sentences. Then say: 'Now I want to show you something.' Project the courtyard drawing annotated with all correct areas. 'This is the courtyard in Kericho. Every formula you have learned — sine area, Heron's, parallelogram, pentagon, hexagon — is written right here on this drawing. This is what you have built, together, over eight lessons. The contractor can now order exactly the right amount of material.' Pause. 'I am going to formally close this Driving Question Board.' [Remove or cross out the driving question card.] 'Every question on this board has been answered. Well done.'	closes the narrative loop of the phenomenon.	
Model Building Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Area Models for Decimal Multiplication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Model Revision (5 minutes): Before the written task, learners individually annotate a personal copy of the original Lesson-1 courtyard drawing (the naive model, with only 'base × height ÷ 2' written on it). They cross out the Lesson-1 formula where it is inapplicable, write the correct formula in each section, and add the computed area from Phase 2. This annotated drawing becomes page 1 of their portfolio submission. Part B — Written Class Activity (15 minutes): Learners individually complete a five-question written task (one question per SLO), administered under assessment conditions. Questions are set in Kericho courtyard or equivalent Kenyan contexts. Q1 (SLO a): Given two sides of a triangular section and the included angle, find its area	Distribute the original Lesson-1 courtyard drawing (naive model copy). Say: 'You have 5 minutes. On this drawing — your Lesson-1 starting point — annotate every section with the correct formula and the area you calculated today. Cross out anything that is wrong. This is the final revision of your model. It belongs in your portfolio.' [WAIT 5 minutes — circulate silently, do NOT intervene.] After 5 minutes: 'Please turn your annotated drawing face down. I am now distributing the written assessment. This is individual work. You have exactly 15 minutes. Show all working for every question — a correct answer with no working earns no marks. Begin.' [Distribute assessment papers. START TIMER.] Circulate silently — do NOT answer mathematical questions, only	Model Comparison (L1 Naive → L8 Expert) / Individual Summative Demonstration. The side-by-side model revision (crossing out the Lesson-1 naive approach and writing the correct formula) makes conceptual growth tangible and personal. It mirrors the scientific practice of revising models in light of new evidence — a key NGSS practice embedded throughout the unit. The five-question individual portfolio task maps directly onto the five SLOs, providing both the teacher and learner with granular diagnostic information about mastery. Requiring working to be shown — not just answers — assesses procedural fluency and conceptual understanding simultaneously.	Summative assessment indicators (marked after lesson): Q1 — correct application of Area = ½ab sin C with correct angle identification (SLO a). Q2 — correct computation of s and correct substitution into Heron's formula with correct arithmetic (SLO b). Q3 — correct formula stated and applied (Area = ab sin θ or base × height with height derived from sin θ) (SLO c). Q4 — correct use of hexagon area formula Area = $(3\sqrt{3}/2)s^2$, correct multiplication by number of tiles (SLO d). Q5 — correct total area, correct cost calculation (area × Ksh 3,200), clear budget comparison with written justification (SLO e). Annotated courtyard drawing — all five formulas correctly placed, computed areas correct to 2 d.p., Lesson-1 errors visibly corrected.

	using the sine formula. Q2 (SLO b): Given three sides of a triangular section, find its area using Heron's Formula. Q3 (SLO c): Given a parallelogram stage with base and included angle, find its area. Q4 (SLO d): Find the area of a regular hexagonal tile with a given side length; find the total border area for a stated number of tiles. Q5 (SLO e — application): The contractor has a budget of Ksh 450,000. Tiles cost Ksh 3,200/m². Using calculated areas, determine whether the budget is sufficient and justify your answer.	clarify task instructions if a learner is genuinely confused about what is being asked. At 10 minutes: 'You have 5 minutes remaining. Check that you have shown all working and written units on every answer.' At 15 minutes: 'Pens down. Please attach your annotated courtyard drawing to your assessment paper and hand both in together.' After collection: 'Let us acknowledge what we have accomplished. You walked into this unit unable to find the area of a triangle without a visible height. You walked out today able to calculate the area of an entire community courtyard — five different polygon types — using five different formulas. The contractor in Kericho can now order the right tiles. That is mathematics in action.'		Use the KICD rubric levels (Exceeds / Meets / Approaches / Below Expectations) to grade both components.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. FORMULA SELECTION MASTERY: During Phase 2 group work, could learners independently select the correct formula for each courtyard section, or did they need prompting? Which formula caused the most confusion — Heron's s-value, the sine-area angle identification, or the hexagon derivation? Plan a targeted 10-minute consolidation activity for the next lesson if more than 25% of learners struggled with formula selection.

2. DQB QUALITY: When learners wrote their one-sentence answer to the driving question in Phase 4, did their sentences explicitly connect the formula (e.g., sine-area, Heron's) to the practical constraint (no perpendicular height measurable)? Or were answers vague? If answers were vague, consider introducing a sentence frame for future unit closures: 'When [constraint], we can still find area by [formula], because [mathematical reason].'

3. PORTFOLIO ASSESSMENT DATA: After marking the five-question written task, record the class distribution across KICD rubric levels for each SLO. Identify any SLO where more than 30% of learners are at 'Below Expectations' — this signals a gap in the unit's instructional sequence that should be addressed at the start of Sub-Strand 2.5.

4. EQUITY OF PARTICIPATION: Review your cold-call records. Did the same 4–5 learners dominate responses across all phases? Identify 2–3 learners who were rarely heard. In Sub-Strand 2.5, use targeted cold-call planning: write their names on a sticky note and deliberately call on them during the first Predict phase.

5. PHENOMENON CONNECTION: Did learners consistently reference the Kericho courtyard — the contractor, the tiles, the material order — during presentations and the DQB closure? Or did their explanations become abstract and de-contextualised? Strong phenomenon connection is the hallmark of effective ARES storyline lessons. If connections were weak, review how the phenomenon was embedded in Lessons 1–3 and strengthen it earlier in the next unit's storyline.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined the complete Kericho community-hall courtyard scale drawing — a floor plan divided into triangular sections (Triangle A, B, C), a parallelogram stage, and a hexagonal tile border — and used all given measurements (side lengths, included angles, and tile side length) to compute the area of every section.
What did I learn?	Learners synthesised all five area-of-polygon formulas from the unit — (a) Area = $\frac{1}{2}ab \sin C$ for triangles with two sides and an included angle; (b) Heron's Formula for triangles with three sides; (c) Area = $ab \sin \theta$ for the parallelogram; (d) Area = $(3\sqrt{3}/2)s^2$ for regular hexagonal tiles — selecting the correct formula for each section and computing a total courtyard area and contractor material cost in Kenyan shillings.
How does this explain the phenomenon?	Learners explained that the Kericho contractor can calculate the exact area of every polygon section — and therefore order precisely the right amount of tiling material — without ever measuring a perpendicular height, because mathematical formulas (sine-area, Heron's, parallelogram, hexagon) allow area to be determined from whichever measurements are physically accessible: side lengths and angles alone are sufficient.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.